

APPLIED PHYSICS REVIEWS

Optical waveguides in lithium niobate: Recent developments and applications

Marco Bazzan^{a)} and Cinzia Sada^{b)}

Dipartimento di Fisica e Astronomia "G. Galilei," Università di Padova, Via Marzolo 8, 35131 Padova, Italy

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The state of the art of optical waveguide fabrication in lithium niobate is reviewed, with particular emphasis on new technologies and recent applications. The attention is mainly devoted to recently developed fabrication methods, such as femtosecond laser writing, ion implantation, and smart cut waveguides as well as to the realization of waveguides with tailored functionalities, such as photorefractive or domain engineered structures. More exotic systems, such as reconfigurable and photorefractive soliton waveguides, are also considered. Classical techniques, such as Ti in-diffusion and proton exchange, are cited and briefly reviewed as a reference standpoint to highlight the recent developments. In all cases, the application-oriented point of view is preferred, in order to provide the reader with an up-to date panorama of the vast possibilities offered by lithium niobate to integrated photonics. © 2015 AIP Publishing LLC. [<http://dx.doi.org/10.1063/1.4931601>]

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^{a)}marco.bazzan@unipd.it

^{b)}cinzia.sada@unipd.it

I. INTRODUCTION

Lithium Niobate (LN) is a key material in photonic industry because of its advantageous combination of functional properties and its commercial availability. These characteristics give the opportunity of using LN as a multi-purpose substrate for the fabrication of integrated optical devices, such as electro and acousto-optical modulators and switches, diffraction gratings, and nonlinear optical frequency converters.

These capabilities were understood since late 60's, pushing the development of methods for the fabrication of integrated optical interconnections on a LN substrate, a basic element for the design of any integrated optical circuit. Well-established techniques, such as local doping by thermal diffusion and proton exchange (PE), were developed since

then, and the present technological achievements of integrated optics in lithium niobate still rely essentially on those methods. In spite of this fact, however, the research for new waveguide fabrication techniques in LN has been continued. On one hand, this was pushed by the demand for better optical performances and, on the other, by some important issues still not solved in the classical waveguide fabrication techniques (among the others, the presence of light-induced charge transport phenomena and the compatibility of the waveguide fabrication technology with ferroelectric domain engineering). This research resulted in alternative technologies bearing the promise to overcome the above mentioned problems and to take advantage of those phenomena that were initially regarded as a drawback, e.g., the use of light-induced phenomena to demonstrate self-written waveguides with the character of an optical soliton.

In this contribution, we review the fundamental aspects of waveguide fabrication and characterization in LN, with a special emphasis on recent developments and opportunities opened up by new understandings on the material inner workings and improved technological achievements.

II. TI IN-DIFFUSION

Since the mid-70's,¹⁻³ the most common integrated optical waveguide-based devices in lithium niobate crystals have been realized by exploiting Ti in-diffusion. For this reason, this technique is one of the most widespread and studied of all the LN technology. Although other techniques allow achieving higher optical confinement, Ti in-diffusion is advantageous for the production of devices since it maintains the electro-optic properties of LiNbO₃ (Refs. 1 and 4-6) and ensures good light confinement on both ordinary and extraordinary refractive indices, allowing both TE and TM propagation modes.⁷ Titanium in-diffusion was exploited together with standard photolithographic techniques to obtain channel waveguides and more complicated optical circuit configurations, such as optical switches, Mach-Zehnder interferometers and optical modulators, couplers with high coupling efficiencies, large bandwidth around 1550 nm, and negligible polarization dependence.⁸⁻¹⁰

A. A short overview of classical Ti in-diffusion process

Titanium is deposited typically by a resistively heated evaporator, E-beam, or sputtering from Ti-target, over the entire substrate. Metal thickness varies from about 0.03 μm to 0.10 μm , depending upon wavelength and the degree of confinement desired. By standard lift-off lithographic techniques, the titanium is selectively removed leaving a metalized pattern corresponding to the desired waveguide pattern. Typical strip widths range from 5 μm (for visible spectral operation) to 8-10 μm for telecom wavelengths at 1.3 μm -1.5 μm . The metal is then diffused into the crystal at temperatures between 980°C and 1050°C for durations between 4 and 12 h (i.e., below the Curie temperature of 1210°C). Many works were performed on the role of the in-diffusion parameters in order to reduce the surface roughness and optimize the optical properties of the material.^{1,11} A key role is played by the annealing atmosphere which has to be

carefully chosen to prevent unwanted lithium out-diffusion. Generally, Ti diffusion into LiNbO₃ is therefore conducted in wet oxygen atmosphere.^{2,12,13}

It is well established that Ti diffuses into LiNbO₃ as a result of cationic migration.^{14,15} In the high temperature oxygen environment, the Ti strip oxidizes forming TiO₂ (Ref. 7) and subsequently, Ti⁴⁺ ions diffuse into LiNbO₃ crystal by occupying vacant Li and Nb sites. A simultaneous inter-diffusion of Li⁺ and Nb⁵⁺ ions into the TiO₂ strip takes place, where Li⁺ ions are highly mobile.¹⁶ Consequently, the reason for annealing in a wet oxygen atmosphere is to prevent the out-diffusion of LiO₂ by neutralizing the Li⁺ ions¹⁷ as a highly stable monolayer of LiOH at the surface.^{18,19} The resulting Ti concentration profile can be derived from the diffusion equation.^{20,21} By exploiting the Fick's law in the case of completely exhausted Ti film, the concentration profile has a semi-Gaussian profile along the depth

$$C(x) = C_S e^{-\frac{x^2}{4Dt}}, \quad (1)$$

where $C_S = \rho\tau/2(\pi Dt)^{1/2}$, x is the depth below the surface, ρ is the number of atoms per unit volume in the deposited film of thickness τ , and D is the diffusion coefficient given by the Arrhenius law $D = D_0 \exp(-Q_D/kT)$, where T is the temperature of the diffusion and Q_D the activation energy. In the range 1000-1100°C, it was reported that $D_0 = 2.19 \times 10^{-4} \text{cm}^2 \text{s}^{-1}$ and $Q_D = 2.18 \text{eV}$.³ The diffusion length $\sigma = \sqrt{4Dt}$ is used to estimate the Ti penetration depth (Figure 1). The Ti doped regions were characterized by a number of techniques, such as secondary-ion mass spectrometry,^{21,22} X-ray micro-analysis (XMA),²³ and X-ray diffraction, as reported in Ref. 19. In the case of deposited Ti strips, the diffusion along the lateral directions can be again modeled using Fick's laws. It should be noted that the diffusion coefficient D depends on the crystallographic direction of the mass flux.¹¹

B. Optical properties of Ti in-diffused waveguides

Some selected measured data of the maximum refractive index jump from different published works are reported in Table I. The data show a certain scattering which is due to the different experimental conditions; therefore, Table I gives an overview of the possible refractive index values that can be obtained with the Ti in-diffusion technique.

Phenomenologically, it was proven that the refractive index jump induced by Ti diffusion-doping on a LN matrix follows a nonlinear dependence on the Ti concentration¹¹

$$\Delta n_{e,o}(z) = A_{e,o}(C_S, \lambda) [C(x)]^{\alpha_{e,o}}, \quad (2)$$

$$A_{e,o}(C_S, \lambda) = \left(\frac{2}{\sqrt{\pi}} C_f \right)^{-\alpha_{e,o}} \left(B_0^{e,o}(\lambda) + B_1^{e,o}(\lambda) \frac{\sqrt{\pi}}{2C_f} C_S \right), \quad (3)$$

$$B_0^e(\lambda) = 0.385 - 0.464\lambda + 0.171\lambda^2, \quad (4)$$

$$B_1^e(\lambda) = 9.13 + 3.85\lambda - 2.49\lambda^2, \quad (5)$$

$$B_0^o(\lambda) = 6.53 \times 10^{-2} - 3.15 \times 10^{-2}\lambda + 7.09 \times 10^{-3}\lambda^2, \quad (6)$$

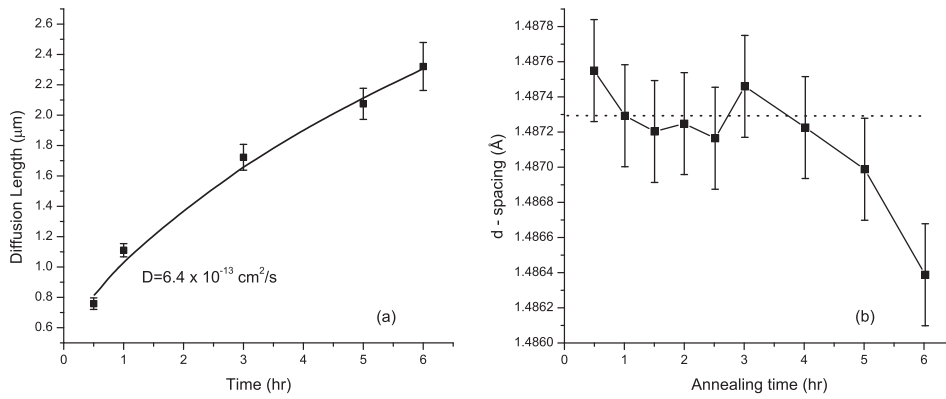


FIG. 1. (a) Diffusion length as a function of the annealing duration. The solid line is the expectation line in the case of $D = 6.4 \times 10^{-13} \text{ cm}^2/\text{s}$. (b) Lattice spacing in the (030) direction as a function of the annealing time. The dashed line refers to the virgin crystal. Reprinted from Ref. 19.

TABLE I. Maximum refractive index jump obtained by Ti in-diffusion in some selected works.

Sample orientation	Film	Thickness (nm)	Atmosphere	Diffusion time (h)	Temperature (°C)	Δn_o	Δn_e	References
Y	Ti	50	Ar (cool in O)	6	960	0.01	0.04	24
Z	Ti	50	Ar (cool in O)	6	960	0.006	0.025	24
Y	TiO ₂	50	Air	10	1000–1100	0.003–0.0019	0.002–0.0018	3
Y	Ti	10–40	Ar (cool in O)	6	950–1100	0.005–0.009	0.007–0.0015	21
Z	Ti	30–210	Ar (cool in O)	9–96	1000–1050	0.001–0.01	0.002–0.02	11

$$B_1^o(\lambda) = 0.478 + 0.464\lambda - 0.348\lambda^2, \quad (7)$$

where C_S and $C(x)$ are defined in (1), λ is expressed in μm , and C_f is the Ti volume concentration in the film prior to diffusion. For extraordinary polarization, α_e is a constant parameter equal to ~ 1.2 for λ between 633 nm and 1550 nm, while for ordinary light, α_o passes from 0.56 to 0.39 in the same spectral range. Calculation of the dispersion characteristics and mode profiles of diffused channel waveguides are generally approximated, using, for example, the effective index method,²⁵ and models were devised which relate mode sizes and propagation constants to diffusion parameters in a simple manner.²⁶ More-exact finite-element calculations, which include dispersion, anisotropy, and the non-linear relation between the refractive index change and Ti-concentration, show good agreement with experimental measurement of mode profiles.²⁷

The microscopic mechanisms of the increase in refractive index in the Ti diffused crystals were investigated in the past and in Refs. 3, 18, and 28. It was calculated that the refractive index changes due to the photo-elastic effect contributed to about half of the observed index changes (about 0.00046 and 0.0014 for the refractive index jump in the ordinary and extraordinary axis, respectively, at 1050 °C of annealing), while the contribution associated with the Ti polarizability is not significant. The refractive index is sensitive also to a change in the spontaneous polarization (a variation of 1% the spontaneous polarization causes an index variations of the order of 10^{-3}). However, a change of the spontaneous polarization causes a lattice strains in the a- and c-axes through the electrostrictive effect, contrary to experimental observations. A very detailed review of the effect of Ti incorporation on the optical properties was made by Syms.²⁹

Ti in-diffused waveguides demonstrated some problem in supporting high power densities achieved in single-mode

guides due to the optical damage, such as crosstalk in switches^{30,31} and drift in interferometers, especially in visible range: optically induced index changes were underlined as a problem at power levels well below $1 \mu\text{W}$ at 633 nm. One possibility is to in-diffuse Ti into MgO-doped lithium niobate⁹ because such doping reduces optical damage in bulk crystals.

C. Recent developments

Recent developments in Ti diffusion technology mainly regarded application of co-doping for further perspectives in telecommunications systems. Special attention was devoted to the development of optical waveguide amplifiers by developing the Er:Ti and Tm:Ti co-doped systems. Erbium diffusion doping of LiNbO₃ was the preferred method to fabricate an excellent laser material for integrated optics. A diode-pumped packaged acousto-optically tunable integrated Ti:Er:LiNbO₃ waveguide laser was presented by Schüfer *et al.* in 1997.³² After Er-doping, single mode channel waveguides were defined by the standard in-diffusion technique of Ti-stripes. By optical pumping at $\lambda = 1480 \text{ nm}$, a wavelength dependent gain of up to 2 dB/cm was achieved ($1530 \text{ nm} < \lambda < 1610 \text{ nm}$): single frequency emission at several wavelengths within the Er-gain band around $\lambda \sim 1550 \text{ nm}$ and an output power of up to 8 mW were reported. Finally, the concept of ring lasers in Er:LiNbO₃ was introduced with their potential to be used as optical gyros. First experimental results were reported by pumping a ring of 60 mm diameter with a laser diode ($\lambda = 1480 \text{ nm}$) clockwise and counter-clockwise. Lasing at $\lambda = 1603 \text{ nm}$ was achieved with an output power up to $150 \mu\text{W}$ (Figure 2).

Tm-doped LN waveguides were designed for the development of optical amplifiers and lasers within the wavelength band $1700 \text{ nm} < \lambda < 1900 \text{ nm}$. A simple 30 mm long Fabry-Perot type Ti:Tm:LN waveguide laser of very low

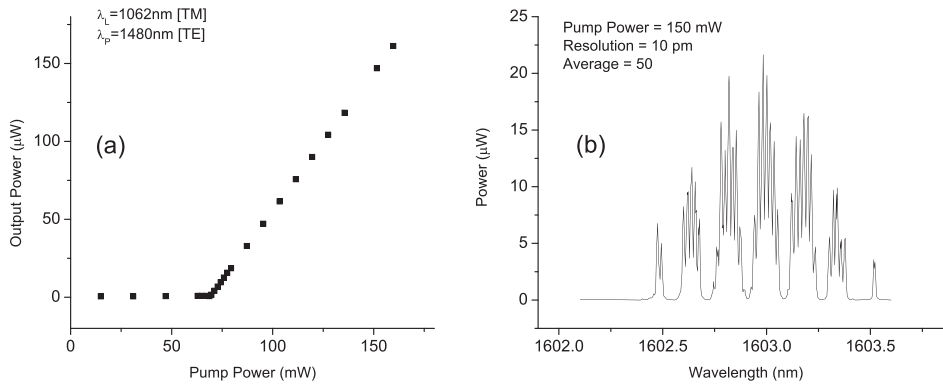


FIG. 2. (a) Power characteristics of the integrated Ti:Er:LiNbO₃ ring laser, TM polarization and (b) emission spectrum of the integrated Ti:Er:LiNbO₃ ring laser, TM polarization. Data reprinted from Ref. 33.

threshold and 1890 nm emission wavelength was proposed.³⁴ Longer Ti:TM:LN waveguides were reported as optical amplifiers and simulation results suggested gain exceeding 30 dB in 90 nm long structures. In particular, laser was demonstrated with emission at 1890 nm. With a laser threshold (1890 nm) at 4 mW coupled pump power, a slope efficiency was found $\approx 13.3\%$.³⁵ Very recently thulium-doped Ti:LN single mode channel waveguides were proposed for applications as photon echo-based quantum memories. Using Ti:TM:LN waveguides, the storage and retrieval of photons ($\lambda = 795$ nm) from entangled photon pairs were demonstrated, thereby temporarily creating entanglement between a photon and a collective atomic excitation.³⁶ Storage of simultaneously arriving, frequency multiplexed photons and recall on demand in the frequency domain were shown as well. These findings have recently attracted great interest in new applications in the field of the quantum information science, since it was exploited in entanglement experiments.³⁷ The feasibility of a two-photon entangled state was demonstrated, where the entangled states are generated and manipulated through electro-optic (EO) modulation.

Ti-in diffusion was also complemented with other techniques. Ridge waveguides were obtained by etching Ti:LiNbO₃ substrates with a mixture of HF and HNO₃ using chromium (Cr) stripes as masks. In this case, a Ti film of 70-nm thickness was vacuum-deposited by e-beam evaporation on the surface and in-diffused at 1060 °C for 8.5 h in argon (7.5 h) and oxygen (1 h) atmosphere, respectively. The resulting Ti concentration profile corresponded to half a Gaussian distribution with a surface concentration of $0.97 \times 10^{21} \text{ cm}^{-3}$ and a depth (FWHM) of 3.24 μm . Low-loss monomode ridge guides with a height of up to 8 μm and a width between 4.5 and 7.0 μm were demonstrated by etching Ti-planar waveguides, with propagation losses in close to 0.3 dB/cm for transverse-electric and 0.9 dB/cm for transverse-magnetic polarization, respectively, at 1.55 μm wavelength.³⁸ In this way, high lateral optical confinement was therefore obtained thanks to the significant refractive index contrast achieved in the ridge waveguide configuration. Some results were also published on the exploitation on a near-stoichiometric LiNbO₃ for applications in the wavelength range from 1.5 μm to mid-IR, where the photorefractive effect is relatively weak. Single-mode near-stoichiometric Ti:LiNbO₃ strip waveguides fabricated by a combination of vapor transport equilibration (VTE) treatment (1060 °C for 12 h) and thermal diffusion of

Ti metal strips were reported.³⁹ Waveguides with optical losses lower than 1.3 dB/cm for TM mode and 1.1 dB/cm for TE mode were obtained with $\Delta n_o \approx 0.01$ and $\Delta n_e \approx 0.034$, respectively.^{9,13,14,19,31,36,40}

III. PROTON EXCHANGE

PE is another classical technique for waveguide fabrication in LN. Compared with Ti in-diffusion, it gives high extraordinary refractive index changes (Δn_e close to 0.1) but is not suited for ordinary light confinement. Often PE is complemented with a subsequent annealing in a controlled atmosphere, leading to the so called Annealed Proton Exchange (APE): This process permits a high polarization rejection and a better resistance to the optical damage, for applications using a shorter wavelength than those exploited in the telecommunications. The annealing, in fact, helps in recovering the nonlinear coefficients that are compromised by standard PE processing. For nonlinear optical applications, Soft Proton Exchange (SPE) and Reverse Proton Exchange waveguides are preferred since they allow a higher refractive index jump and therefore a more efficient optical confinement. These techniques maintain unperturbed the electro-optics and nonlinear properties of the crystal at the cost of a lower index increase Δn_e (between 0.01 and 0.03). Vapor-phase Proton Exchange (VPE) has been also proposed to combine a strong confinement ($\Delta n_e = 0.09$) with preserved nonlinearities⁴¹ although the control of the fabrication conditions is rather complicated.

A. Experimental preparation process

Proton exchange is a quite straightforward technique, consisting in the immersion of a lithium niobate specimen into a liquid source of hydrogen at high temperature. Under these conditions, the replacement of lithium ions (Li⁺) by hydrogen ions, or protons (H⁺), is promoted. This replacement causes a change in extraordinary refractive index, thus forming a waveguide supporting modes only on the extraordinary polarization. The most common sources of Hydrogen ions H⁺ are the benzoic acid (BA) and the toluic acid heated at temperatures ranging from 150 to 400 °C. When the crystal is immersed in such a bath for several hours, Li⁺ out-diffuses from the crystal matrix and is replaced by H⁺ that in-diffuses into the crystal, so that mass transfer occurs between the substrate and the melt. The process is performed

by exploiting sealed ampoules. The exchange time can vary from several hours to several days depending on the aimed waveguide depth and the melt composition. When the process is complete, the container is turned upside down, allowed to cool and then opened to get out the samples. The process details were described in details by Korkishko and Fedorov and will not be further described here.⁴² The process dynamics is well described by the diffusion equations with the constraints of local charge balance given by the ion replacement of isovalent species^{42,43}

$$\frac{\partial C_H}{\partial t} = \frac{\partial}{\partial y} \left[D_{Li-H} \left(\frac{\partial C_H}{\partial y} \right) \right], \quad (8)$$

with $D_{Li-H} = \frac{D_{H,s}D_{Li,s}}{D_{H,s}C_{H+} + D_{Li,s}C_{Li}}(C_{Li} + C_H)$, where C_H is the hydrogen concentration and C_{Li} the lithium one. The coefficients $D_{H,s}$ and $D_{Li,s}$ are connected with the self-diffusion coefficients D_H and D_{Li} , respectively.⁴² The process leads to a compositional and structural modification in the exchanged region, which is generically referred to as $Li_{1-x}H_xNbO_3$ layer, showing a complex phase behavior. Up to seven different crystallographic phases (namely, α , κ_1 , κ_2 , β_1 , β_2 , β_3 , and β_4) can be formed, depending on the hydrogen concentration x , the crystallographic orientation, and the rate

of substitution of Li^+ by H^+ (x), respectively. In particular, the self-diffusion coefficients differ depending on the $H_xLi_{1-x}NbO_3$ phase type.^{42,44,45} Each phase originates as an individual sub-layer depending on the local composition x . In every single one of them, the lattice mismatch perpendicular to the surface changes abruptly and Δn_e is a linear function of the local proton concentration. To optimize the optical properties of the proton exchange waveguides, it was also proposed to buffer the BA melt with lithium benzoate (LB), a technique often referred to as soft proton exchange.⁴²

The presence of LB, in fact, slows down the proton exchange rate and prevents complete depletion of the lithium from the surface of the $LiNbO_3$. In general, phase modulators fabricated by SPE technique exhibited a better electro-optical efficiency compared to the $LiNbO_3$ modulators produced by the standard and improved APE techniques and therefore SPE is preferred.⁴⁶ Each phase in the $H_xLi_{1-x}NbO_3$ is characterized by different structural and optical properties, as reported in Table II.^{42,45,47} As a summary, the dependence of the phase formation on the temperature and different hydrogen content is reported in Figure 3 as published in Ref. 42, while the hydrogen concentration profile in the exchanged regions in function of the crystallographic phase formed in that region is presented in Figure 4.

TABLE II. Optical and structural characteristics of proton exchange waveguides in lithium niobate crystals.⁴² BA, benzoic acid; LB, lithium benzoate; PA, pyrophosphoric acid.

Cut	PE source	Annealing temperature	$H_xLi_{1-x}NbO_3$ phase	Δn_e	x
Tab.1Z	BA	...	β_2	...	0.4–0.5
		275 °C	β_1	0.11	0.32
X	BA	...	β_2	0.12	0.85 ^a
Z	BA	400 °C	κ_2^{HT} to κ_1^{HT}	...	<0.22
Z	BA	...	β_2	...	0.29
		400 °C	κ_2^{HT} to κ_1^{HT}	...	...
Z	BA	275 °C	β_1	0.11	0.32
		400 °C	κ_1^{HT}	0.047	0.14
Z, X	BA	400 °C	from β_2 to κ_1^{HT}	...	...
Z, X	BA	...	β_2	...	0.65–0.75
Z	BA	...	β_2	...	0.75
Z	BA	...	β_2	...	0.33
X	BA + 1% LB	300 °C	κ_2^{LT}	0.081	0.315
		320 °C	κ_1^{LT}	0.039	0.315
		340 °C	α or κ_1^{LT}	0.039	0.275
		400 °C	κ_1^{HT}	0.0035	0.020
Z	BA	300–370 °C	from β_2 to α	...	...
Z	PA	...	β_4	0.68	0.132
		200	β_4 or β_3	0.68	0.124
		260	β_1	0.65	0.114
		280	β_1	0.64	0.113
		300	β_1	0.55	0.110
		320	β_1	0.55	0.104
		340	κ_2^{LT}	0.51	0.099
		360	$\kappa_2^{LT} + \kappa_2^{HT}$	0.41	0.092
Z	PA	...	β_4	0.148	0.75
		250 °C	β_3 or β_1	0.12	0.66
Z	PA	...	β_4	>0.14	0.9
		300 °C	β_3 or β_1	...	0.5
		300 °C	β_1	...	0.4

^aOnly Li concentration has been determined.

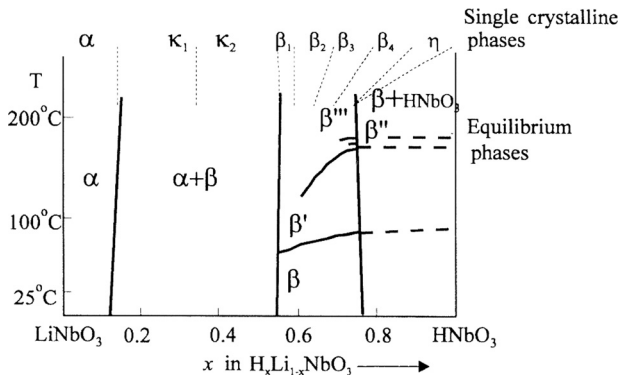


FIG. 3. Equilibrium and stressed phase diagrams of $H_xLi_{1-x}NbO_3$. Reproduced with permission from Yu. N. Korkishko and V. A. Fedorov, IEEE J. Sel. Top. Quantum Electron. 2(2), 187–196 (1996). Copyright 1996 Institute of Electrical and Electronic Engineers.

Waveguides fabricated from pure melts are characterized by a step refractive index profile but shows degraded electro-optic coefficient, large scattering and insertion loss, and refractive index instabilities. It is therefore a common practice to anneal the samples after the exchange process in order to eliminate the compositional instabilities and to restore the material properties. Annealing, in fact, helps avoiding problems related to a high fraction of hydrogen ions in the exchanged waveguides: this process is known as Annealed Proton Exchange (APE). The post-exchange annealing process corresponds to a different refractive distribution of the refractive index, generally described by the following equation:

$$\Delta n_e = \Delta n_{e,0} e^{-\left(\frac{y}{D_V}\right)^a} \left\{ \frac{1}{2} \operatorname{erf} \left[\frac{w}{2D_x} \left(1 + \frac{2x}{w} \right) \right] + \frac{1}{2} \operatorname{erf} \left[\frac{w}{2D_x} \left(1 - \frac{2x}{w} \right) \right] \right\}^b, \quad (9)$$

where $\Delta n_{e,0}$ is the maximum refractive index change on the surface, D_x is the hydrogen diffusion depth along the direction x that depends on the exchange temperature via the Arrhenius law, w is the lateral dimension of the waveguide, while y is

the in-depth direction. The parameter a ranges between 10 and 23 (Refs. 48 and 49) and b takes into account the effect of the annealing and relative further diffusion of Hydrogen in the matrix. D_V , instead, is the diffusion depth after annealing or effective guide depth: $D_V = \sqrt{tD}(1 + bt_a^c)$. D_V is strongly dependent on the post-exchange time t_a and the post-exchange temperature T_a , $b = 0.7031 \text{ h}^{-c}$ for $T_a = 300^\circ\text{C}$ and increases up to 1.2884 for $T_a = 400^\circ\text{C}$. The parameter c instead was found to be equal to 0.0754 for $T = 300^\circ\text{C}$ and to 0.7577 for $T_a = 400^\circ\text{C}$, respectively.

It is worth mentioning that Korkishko and co-workers demonstrated also the feasibility of the high-temperature proton exchange (HTPE) to fabricate high-quality $LiNbO_3$ optical waveguides exploiting the stearic acid diluted by lithium stearate as well. In this case, the alpha-phase produced presented the same crystalline structure as that of pure $LiNbO_3$ crystal, and maintained the excellent nonlinear and electro-optical properties of the bulk material.⁵⁰ Recently, SPE process was again deeply investigated to obtain more stable waveguides and higher figures of merit. In Ref. 51, it is proposed a process with 0–4% of lithium benzoate in benzoic acid combined to a reverse proton exchange for 24 h at 300°C in a eutectic solution of $LiNO_3$ (30%), $NaNO_3$ (17.8%), and KNO_3 (52.2%). In Ref. 52, the so-called High Index Soft Proton Exchange (HISoPE) process is proposed, which uses as proton sources LB in the range 2–2.4%, i.e., lower than the threshold concentration $\rho_{th} = 2.7\%$ above which SPE waveguides at 300°C are obtained. Finally, in order to get even ordinary refractive index jump, it was proposed to perform⁵³ a reverse exchange in a melt enriched in lithium on a substrate previously exchanged in normal benzoic acid. Layers supporting several modes with ordinary polarization were therefore obtained, with losses better than 0.7 dB/cm.

B. Recent results and applications

Proton exchange was in general used for getting high performance optical waveguides and different devices were

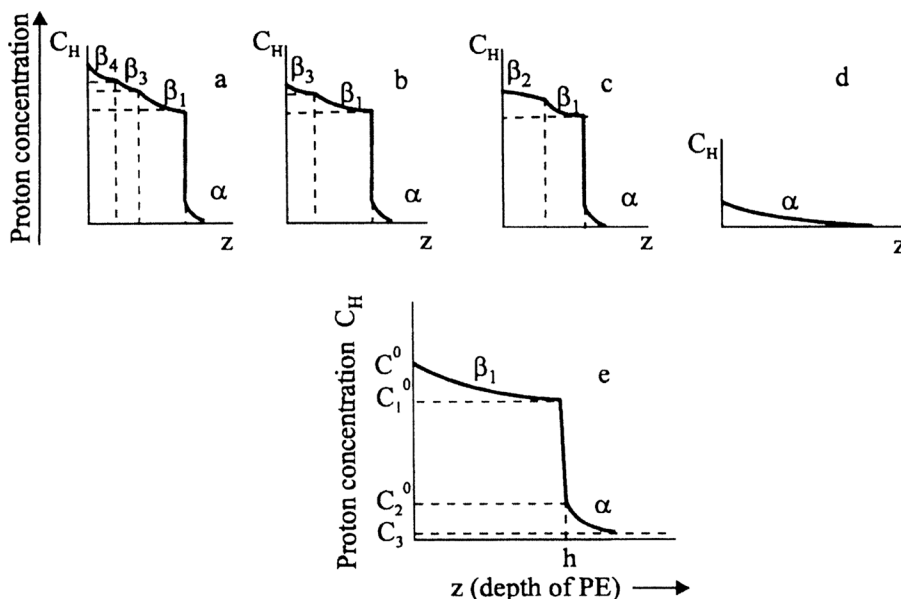


FIG. 4. Proton diffusion depending on acidity of PE sources. (a) β_4 - $H_xLi_{1-x}NbO_3$ phase formed at surface (PE in pyrophosphoric acid); (b) β_3 -phase (glycerine + $KHSO_4$); (c) β_2 -phase (benzoic acid); (d) α -phase (BA + more than 3% LB), and (e) β_1 -phase (BA + 0.5 to 2.5% LB). Reproduced with permission from Yu. Korkishko and V. A. Fedorov, “Ion exchange process in $LiNbO_3$,” in *Ion Exchange in Single Crystals for Integrated Optics and Optoelectronics* (Cambridge International Science Publishing, Cambridge, 1999), pp. 97–261. Copyright 1999 Cambridge International Science Publishing.

already demonstrated in the years 1990–2005, such as modulators,⁵⁴ switches,⁵⁵ multiplexers, Y branches.⁵⁶ In general, proton exchange is preferred for any applications, where propagation in the visible spectrum is necessary and when the limitation to extraordinary polarization is not detrimental. The fact that only extraordinary refractive index variations can be achieved, on the other hand, was used also for polarization control.⁵⁷ Special application of proton exchange was published on photorefractive-resistant substrates, such as MgO:LiNbO₃ (Refs. 58, 59, 60), even combined with Ti indiffusion⁶¹ and in periodically poled structures (see Section VIII). Many reviews on this topic were proposed and will not be further discussed here, addressing the reader to excellent contributes in the literature.^{10,42,62,63}

Focusing the attention in the last ten years, great efforts were performed in optimizing the process to guarantee high electro-optic coefficients and strongly improve the optical coupling by decreasing the optical losses. Waveguide tapers with coupling losses lowered up to 0.7 dB (Ref. 64) realized in LiNbO₃ using the soft proton exchange and the waveguide segmentation techniques were reported. Integrated wavelength de-multiplexer produced by double proton exchange on z-cut lithium niobate was also demonstrated.⁶⁵ Finally, it was reported the feasibility of very high quality channel waveguides with propagation losses as low as 0.086 dB/cm and a coupling efficiency to optical fibers of 90% at 1550 nm. The splitting ratio of a set of directional couplers could be predicted with an accuracy of ± 0.06 .⁶⁶ Recently, the combination of PE to other technique was explored to get new configurations. Proton-exchanged LiNbO₃ crystals were subjected to 70-MeV argon-ion irradiation. It was found that the lattice disorder induced by the proton exchange process was partially recovered, the proton-exchanged layer was broadened and the swift argon-ion irradiation induced an increase in extraordinary refractive index and formed another waveguide structure beneath the proton-exchanged waveguide.⁶⁷

IV. ION IMPLANTATION

Ion implantation is a well-known technique to prepare optical waveguides in integrated photonics and modern communication systems. It consists in bombarding a surface at a given angle with respect to the surface normal with ions at an impact energy E_0 so that, after releasing their energy, the impinging ions stop at an average depth from the surface, the so-called projected range, R_p . In this process, the energy of the bombarding ion beam is transferred to the crystal lattice by way of two main processes dependent on the ions velocity: energy loss due to electronic excitations and the nuclear collision losses. The first contribute is quantified by the electronic stopping power $S_e = (dE/dx)_e$ and dominates at higher velocities (kinetic energy typically higher than 8–10 KeV/amu). In this regime, the bombarding ion's trajectory in the material is almost straight, being not affected by collisions with the material constituents. At lower velocity, the nuclear collisions are significant and the ion path is consequently characterized by abrupt changes of direction. In this case, the

energy loss is quantified by the nuclear stopping power $S_n = (dE/dx)_n$. Because of the energy and momentum transfer during the nuclear collisions, the so-called displacement damage is induced, resulting in the formation of point and/or extended defects. Moreover, both the electronic and nuclear processes tend to create color centers (optical absorption sites). Consequently, an annealing treatment is commonly necessary for all the ion-implanted waveguides in crystals in order to recover the optical properties of the material.⁶⁸

The energy release and its effect on the material can be studied in detail with accurate simulations to trace the ion track within the material. Most often, Monte Carlo approach is exploited in the approximation of binary collisions.⁶⁹ Recently, even molecular dynamics based on *ab-initio* calculations was carried out focusing on Ehrenfest dynamics and time-dependent density functional theory to calculate electronic stopping power S_e of energetic ions in graphitic targets, from first principles.⁷⁰ A detailed description of the physical phenomena underlying the ion implantation was deeply reviewed in the past,^{71,72} as well as of the instrumental set-ups.^{73,74}

By exploiting standard beam rastering techniques, a homogenous irradiation can be obtained all over the sample surface, with ion beam diameters in the range of 0.002–0.01 cm. When combined with photolithographic patterning of the surface, this procedure allows for selective implantation in 2D patterns.⁶⁸ Alternatively, the ion beam can be focused down to about 100 nm (Focused Ion Beam, FIB) and used to perform selective implantation to spatially defined regions.⁷⁵ In the case of lithium niobate crystals, both approaches were used to create optical waveguides.⁷⁶

A. Effects of ion implantation on the optical properties

Depending on the interactions between the bombarding ions and the material, i.e., on the impact of the electronic energy loss and the nuclear one and their eventual interplay, different optical responses can be obtained:

- (1) At the end of range region, where the ions stop their trajectories, the nuclear stopping power plays a key role. The collisions between the implanted species and the matrix atoms produce displacement damage. As the material evolves towards an “amorphous” state, the ordinary refractive index tends monotonically to decrease with the implantation fluence, while the extraordinary first increases, reaching a maximum value and then decreases.
- (2) Along the implantation track, the refractive index can undergo modifications correlated with the electronic energy loss. The term “ion irradiation” was introduced by Olivares and co-workers⁷⁷ to distinguish this process with respect to the standard ion implantation in the matrix. In fact, depending on the implantation conditions, different regimes can be exploited: (i) implantation at high energy (4–100 MeV) and with heavy ions ($A > 12$) so that electronic stopping power dominates and a complete amorphization of the lattice can be obtained if the electronic stopping power exceeds a certain threshold;^{75,78,79} (ii) an intermediate regime where the

electronic energy loss dominates but the contribution of the nuclear energy loss cannot be neglected.

In the following, the different processes will be discussed with reference to their impact on the optical properties and to waveguide fabrication.

1. Optical waveguides: Nuclear energy loss contribution

It is well known that at the end of the ion track, it takes place a more or less developed amorphization process induced by nuclear energy deposition.^{71,76,80} The link between the defects created by the ion implantation and the waveguiding properties of the implanted layer has been under debate for a long time.^{81–83} Different contributes induced by the implantation have to be taken into account: (i) the modification in the material spontaneous polarization P_s (that decreases towards zero in the amorphous region); (ii) the strain developed in the implanted region which contributes *via* the photoelastic tensor; (iii) the changes due to the modifications in the electronic density, which modify the average refractive index according to the Lorenz–Lorentz equation. All these contributes are related to the lattice damage ratio k , defined as the fraction of displaced atom $N_D(x)$ over the total atoms contained in an arbitrarily small volume at the depth x from the surface: $k = 0$ corresponds to the virgin crystal, while $k = 1$ indicates a complete amorphisation (generally, k is measured by Rutherford Backscattering in channeling mode, using the Nb signal). By assuming a linear dependence of the relevant quantities upon k , the following equation can be obtained:^{83–87}

$$\begin{aligned}\Delta n_{ij} &= \Delta n_{ij}^{P_s}(k) + \Delta n_{ij}^S(k) + \Delta n_{ij}^M(k), \\ \Delta n_{ij}^{P_s}(k) &= \frac{1}{2} n_{ij,0}^3 (1-k)^2 g_{ij3} P_s^2, \\ \Delta n_{ij}^S(k) &= \frac{1}{2} n_{ij,0}^3 (1-k) p_{ijk} \frac{\Delta V}{V}, \\ \Delta n_{ij}^M(k) &= \frac{(n^2 - 1)(n^2 + 2)}{6n} \\ &\quad \times \left[\ln \left(1 + \frac{\Delta \alpha_{M,i}}{\alpha_{M,i}} k \right) + \ln \left(1 + \frac{\Delta V_M}{V_M} k \right) \right],\end{aligned}\quad (10)$$

where the subscript 0 refers to the absence of damage and i, j label to the crystallographic directions. g_{ij3} is the quadratic electro-optic tensor, p_{ijk} is the photoelastic tensor, $\frac{\Delta \alpha_{M,i}}{\alpha_{M,i}}$ and

$\frac{\Delta V_M}{V_M}$ are, respectively, the relative changes in the molar polarizability and in the molar volume passing from a virgin crystal to a fully amorphized one ($k = 1$). $n = \sqrt[3]{n_o^2 n_e}$ is the average LN refractive index.

In Figure 5, the ordinary and extraordinary refractive indices as a function of the lattice damage are reported as published in Ref. 83 in X and Z cut lithium niobate crystals.

The models account for the decrease in the ordinary refractive index with the lattice damage and foresee an increase in the extraordinary refractive index as far as the k value is less than 0.2–0.3 in X and Y cut crystals and less than 0.3–0.4 for Z-cut ones, respectively. According to this view, an optical waveguide can be created in the implanted region either because the decrease in the refractive index at the end of range region or because of the increase in the extraordinary index in the region with moderate damage ratio. In the first case, an optical barrier is created at the end of range, and the light is confined between this barrier and the crystal surface.^{68,88} In the second case, by properly adjusting the implantation conditions, a step-index optical guide can be obtained.⁸⁹ This model was exploited to fit the refractive index profiles in O-implanted lithium niobate crystals, Z-cut, with fluences in the range $5\text{--}40 \times 10^{14}$ ions/cm², energies laying between 0.55 and 4.0 MeV, and post annealed at 260° for 30 min, respectively. In this case, optical waveguides supporting at least 1 mode at $\lambda = 632.8$ nm depending on the process parameters were reported, with optical loss of the order of 1 dB/cm. Excellent agreements were reported also in Si implantation, where monomode optical waveguides were therefore obtained by exploiting implanted energies of 500 KeV (Ref. 90) and 3 MeV (Ref. 91), respectively.

2. Optical waveguides: Electronic energy loss contribute

In the last decade, great efforts were also dedicated to the understanding of the effect of the electronic energy loss mechanism as well, since it was discovered to affect the material structure and, consequently, its refractive index. In Ref. 75, this topic was already reviewed in great detail, with the state of the art up to 2011. In the following, only the most import issues will be underlined. The effect of ion-beam damage in the electronic stopping regime for many oxide, semiconductor, and even metallic materials, is the formation of highly-defective linear tracks along the trajectory of single ions with a typical diameter in the nanoscale,

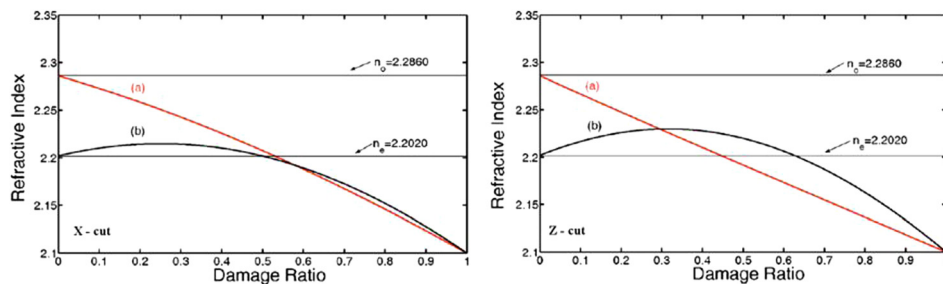


FIG. 5. Ordinary (n_o) and extraordinary (n_e) refractive index variations as a function of the lattice damage ratio k as computed by the model of Ref. 83 (the arrows indicate the values expected in virgin samples): (Left) X-cut. (Right) Z-cut. Reproduced with permission from J.-J. Yin *et al.*, J. Appl. Phys. **108**, 033105 (2010). Copyright 2010 American Institute of Physics Publishing.

provided that the electronic stopping power S_e exceeds an intrinsic threshold value S_{th} .^{78,92–94} These tracks are made of a central amorphous core which extends to a certain depth from the surface, down to several micrometers, and a surrounding defective halo containing point and extended defects, as revealed by transmission electron microscopy (TEM)^{95,96} and optical techniques.⁷⁷ The threshold value was evaluated in different experiments and was found to be in the range between 5 and 10 KeV/nm in the case of LiNbO₃,^{94,97} well below what happens in Si (35–40 KeV/nm). Above the threshold, the radius of the tracks increases monotonically with the incident stopping power.⁹⁸ If $S_e > S_{th}$, for sufficiently high doses, random tracks begin to overlap and constitute a homogeneous defective (amorphous) layer. Depending on the stopping power curve, one may form this layer at the surface or buried into the material. Therefore, a strong research activity was devoted to the fabrication of optical waveguides with ions of medium mass and high energy (swift heavy ions or SHI), where an amorphous region is produced by electronic excitation processes for electronic stopping powers above the threshold value.^{99,100} This regime allows the separation of the region of electronic excitations from that associated with nuclear damage and implantation. Its location and width can be suitably controlled by the irradiation fluence and it can be well separated from the deeper end of range region.¹⁰¹ Moreover, the refractive index contrast reaches higher values than for conventional implanted guides and the boundary between waveguiding and barrier layers is sharper. Due to all these advantages, there has recently been a considerable increase in the number of research works pushing the use of electronic excitation as a novel means to fabricate optical waveguides and other integrated optical devices.^{102–105}

3. Recovering of the optical properties after the ion implantation

The aim of annealing is to remove the detrimental defects in the waveguide preserving the refractive index contrast and the electro-optical or non-linear optical properties of the material. Commonly conventional annealing furnaces,¹⁰⁶ rapid thermal annealings,¹⁰⁷ and laser beam processing¹⁰⁸ were exploited to reach this goal. In the case of thermal treatments, they are performed in the range 200–500 °C for duration lasting from tens of minutes to several hours, leading to a reduction in the propagation losses of the waveguides as low as 0.1 dB/cm, accompanied, however

to a reduction of the refractive index contrast.¹⁰⁹ Recently, a systematic investigation on the role of the annealing was performed by Montanari and co-workers in the case of C multi-energy implantation (0.35–3.9 MeV; fluences in the range $0.363\text{--}1.325 \times 10^{14} \text{ cm}^{-2}$) to get the step-index profile, with waveguide thicknesses close to 3 μm .⁸⁹ The annealing process was performed at temperature from 100 °C to 415 °C for 30 min, respectively, demonstrating that the birefringence constantly recovers as the annealing temperature increases (Figure 6(a)). The annealed samples at 280 °C showed the optimal characteristics in terms of refractive index profile and material attenuation. In the best case, the absorption coefficient was lower than 0.8 dB/cm, with an optical transmission spectrum that is close to that of the un-implanted sample (Figure 6(b)).

The rapid thermal annealing was instead preferred to prevent undesired thermal diffusion.¹⁰⁷ Recently, laser beam processing was presented as a valid alternative to get a local thermal treatment by exploiting the absorbed radiation at the surface and, therefore, preventing re-growth or solid phase epitaxy.^{108,110}

4. 1D and 2D geometries

Planar waveguides may be fabricated by uniformly implanting the sample surface, in which case no specific masks are required. However, for 2D waveguide formation, further technological steps are needed to define the geometry of the implanted region. In the earlier works, masks were obtained by depositing a thin photoresist or metal (e.g., Au) film, and using standard photolithography technique.^{68,109} In this case, the thickness of the mask was chosen so that the incoming ion underwent to energy loss while crossing the mask deposit and could stop at the correct depth, generating a decrease in the refractive index that can act as side wall optical barrier. Meanwhile, the incoming ions passing in the unmasked region, stop at a depth suitable to realize an optical barrier at the bottom of the waveguide.^{111,112} In this configuration, channel waveguides and Mach-Zehnder modulators were obtained by implanting carbon at 3.9 MeV with fluences in the range $2\text{--}4 \times 10^{18} \text{ m}^{-2}$. Alternatively, the common way to proceed and get a better lateral confinement starts from an implantation in a planar waveguide configuration, followed by a photolithographic step to cover those regions where the channel waveguide should be obtained. After masking, a series of successive implantation can be performed with tailored energy to build up index-decreased sidewalls: In this way, the

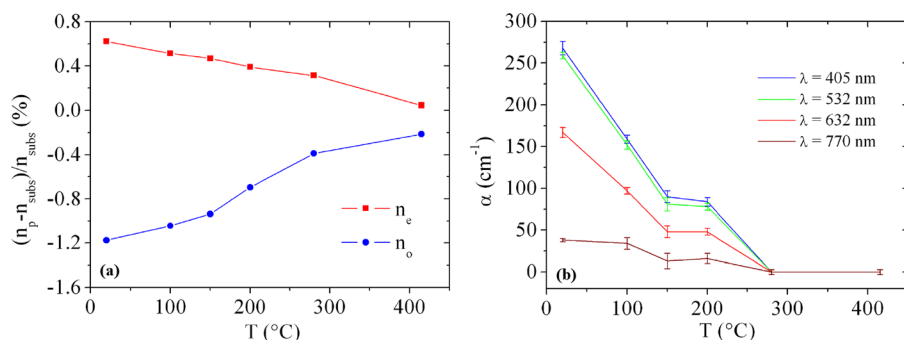


FIG. 6. (a) Variation of n_e and n_o depending on the annealing temperature; (b) absorption coefficient of implanted C-implanted LiNbO₃ depending on the annealing. Reproduced with permission from G. B. Montanari *et al.*, Opt. Express **20**, 4444 (2012). Copyright 2012 Optical Society of America Publishing.

waveguides are formed in the mask-protected regions, being confined the bottom barrier (of the planar waveguide implantation) and sidewalls, together. Moretti *et al.* suggested mounting the sample on a goniometric holder and implanting it through a movable metal slit: By adjusting the incoming ion angle with respect to the surface, it was demonstrated that the implanted ions could create a closed barrier wall or an increased refractive index well in the substrate, therefore allowing for channel waveguide realization.¹¹³

B. Fabrication examples and typical performances

A summary of the published results of ion implantation in lithium niobate crystal was recently reviewed by Chen *et al.*¹⁰⁹ and Pena-Rodriguez⁷⁵ referring on the state of the art updated in 2011. The majority of the published works was focused on the fabrication and refractive index characterization of the LiNbO₃ waveguides by implantation of ions at low fluence (10^{14} ions/cm²). Many older works were published on the He⁺-implantation in LiNbO₃: waveguides at high fluences (10^{16} ions/cm²) were obtained in which the ordinary refractive index had a typical barrier-type profile, whilst the extraordinary refractive index showed more complex changes: At the surface, n_e was increased, but the maximum positive change was found to be peaked deep inside the guide, resulting in so-called “missing modes,” and an index-decreased barrier was located at the end of ion track.¹⁰⁹ This is the typical case where the electronic energy loss is significant but the nuclear energy loss must be considered as well. In any case, both the optical barrier and the positive index well contribute to the light confinement. It is worth mentioning that while He⁺ ion implantation generates large-dimension clusters of different crystalline nature,¹¹¹ proton implantation gives only point defects, even at much higher fluences (up to 2×10^{17} ions/cm²). In this case, the implanted region could approximately recover to the original lattice. Implantation of other ions at low fluences, such as B, C, O, F, Si, and P, n_e was found to have similar behaviors in the guide regions.¹⁰⁹ However in the case of O and C implantation, it was also reported the presence of tensile strain that may contribute to the decrease in the ordinary refractive index variation through the photo-elastic effect.¹¹⁴ By irradiating LiNbO₃ with F ions at moderate fluences (2×10^{14} cm⁻²), waveguides with nonlinear properties comparable to those of other LiNbO₃ guides were obtained, having strong confinement and losses around 1 dB/cm. More recently, planar waveguides with a step-like index profile were obtained by 20 and 22 MeV F ion irradiation into LiNbO₃, under experimental conditions that enhance the contribution of the electronic energy loss.⁷⁵ Waveguides in co-doped Nd (0.3 At.%):MgO (5 mol.%) X-cut LiNbO₃ crystals have also been obtained by Chen and Hu coworkers using O ions.^{90,91} Ion implantation of Si (Ref. 91) was exploited as well, realizing monomode optical waveguide and indicating the potential generality of the method. Of particular interest are the waveguides obtained by irradiating in the regime of high electronic stopping power ($\sim 5\text{--}6$ keV/nm) and ultra-low fluence (2×10^{12} cm⁻²),¹¹⁵ because of the short irradiation time required.

The electro-optical properties of the light ion-implanted LiNbO₃ waveguides were investigated by measuring the angular shift of the guided mode when an electrical field was applied. In the case of H⁺ implantation, the values of the electro-optic coefficients (r_{13} and r_{33}) were well-preserved in the waveguides, independently of the implanted-proton fluence in the investigated range ($1\text{--}10 \times 10^{16}$ ions/cm²). Also, the non-linear d_{33} coefficients had almost the same value of virgin LiNbO₃. In the case of He⁺-implanted LiNbO₃ waveguides, instead, the r_{13} was reduced by 20% and the non-linear d_{33} coefficients were reduced by 48%.¹⁰⁹ Pena-Rodriguez *et al.* studied the second-harmonic generation (SHG) of 20 and 22 MeV F ion-implanted LiNbO₃ waveguides, and high SHG coefficients were obtained (50–80% of a bulk LiNbO₃).⁷⁵ An interesting phenomenon was observed in the optical barrier at the end of range: a strongly enhanced SHG. Moreover, implantation of LiNbO₃ with MeV metal ions could be used to prepare colloidal nanoclusters with great potential in a variety of device applications. The non-linear refractive index of 1.5 MeV Ag⁺-implanted LiNbO₃ waveguides was positive with a magnitude of approximately 5×10^{10} cm² W at 532 nm, which compared well to the best known third-order non-linear optical materials.^{116,117}

V. LASER-WRITTEN WAVEGUIDES

It has been demonstrated that in LN optical waveguides can be obtained by scanning a crystal with a focused laser beam. During the 90's, low-power cw lasers were used, taking advantage of the particular form of the refractive index changes induced by the photorefractive effect. More recently, a similar approach was demonstrated, making use of modern ultra-short laser pulses. Even if the two techniques present a number of analogies, such as the setup scheme, the dependence on the crystal orientation, and the obtainable structures, they actually depend on different physical mechanisms. The main advantage of this approach is that three-dimensional waveguide structures can be fabricated in the bulk of a sample without any further processing and was proven to be useful in a number of practical applications.

A. Photorefractive laser-written waveguides

As shown in Figure 7(a), a focused light beam propagating perpendicular to the optical axis of a LN crystal can give rise to a complex refractive index pattern. This pattern consists in a negative refractive index variation at the core of the beam and in a two-sided lobe structure disposed along the c direction with a positive refractive index change.¹¹⁸ Itoh *et al.* demonstrated that each one of these lobes with positive index can be used to guide light.¹¹⁹

By using computer-controlled motion, straight and curved waveguides can be realized in a full 3D geometry (Figure 7(b)). In the best conditions (70 mW of pump power, focused by an optical system with 0.1 NA at a wavelength of 514 nm, writing velocity of 0.1 mm/s), a positive refractive index jump of more than 10^{-3} for an extraordinarily polarized guided beam can be obtained, which was sufficient to confine light even in presence of moderate waveguide

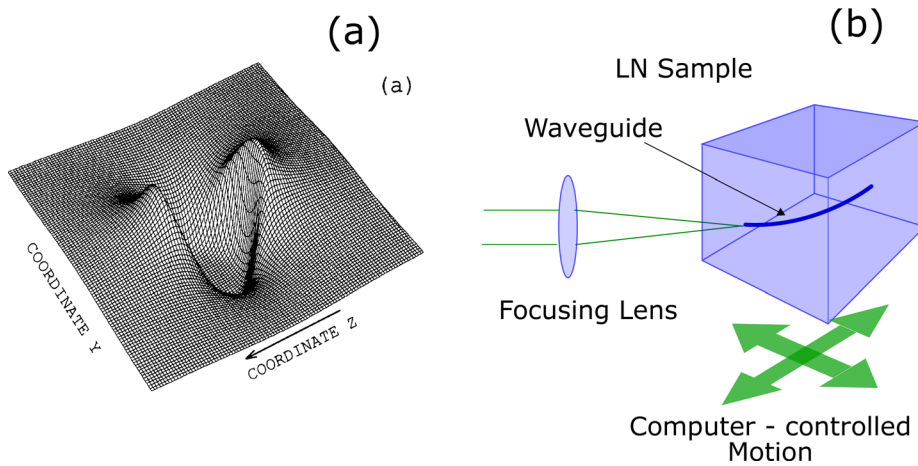


FIG. 7. (a) Calculated refractive index change induced by a focused beam. The two lobes with positive refractive index changes are clearly visible (reproduced with permission from A. A. Zozulya and D. Z. Anderson, Phys. Rev. A **51**(2), 1520 (1995). Copyright 1995 The American Physical Society). (b) Scheme of the focused photorefractive beam writing technique.

bending.¹²⁰ The produced waveguides were single mode and were able to transmit up to 40% of the injected beam power, taking into account both propagation and coupling losses. The mode field diameter was about $7\ \mu\text{m}$, but obviously could be changed using a different focusing optics. As this process is based on the photorefractive effect, the phenomenon is effective only if the crystal optical axis is close to be perpendicular to the direction of the written line and for an extraordinary reading beam.

B. Femto-second laser-written waveguides

The illumination of LN with ultrashort laser pulses results in structural modifications which can be used to fabricate optical waveguides.^{121,122} Depending on the processing parameters like pulse energy, pulse duration, or numerical aperture of the focusing lens, two types of laser induced modification can be distinguished, depending on the pulse energy E .¹²³

Type I: In x-cut crystals, for $E < 0.5\ \mu\text{J}$ with the structured lines along the crystal y-direction, a localized refractive index increase is observed at the laser focus position with no visible material damage. The writing laser may be linearly polarized either parallel or perpendicular to the waveguide. As for the photorefractive laser writing technique, if the waveguide axis is directed along the z-direction, no such index change is observed. For pulse energies above $E > 0.5\ \mu\text{J}$, instead, optical damage leading to a refractive index decrease was generated at the laser focus, but positive index change was still observed at a position of reduced intensity close to, but not at the focus. The positive index change was detected only for n_e , i.e., for light polarized along the z-direction of the crystal, with magnitudes of $\sim 5 \times 10^{-4}$, whereas the index decrease was visible in both polarizations. In Z-cut samples, the modifications were insensitive to the direction of the waveguide and the polarization of the writing laser. Loss measurements of type I waveguides indicated high propagation losses in the region of $9.0\ \text{dB cm}^{-1}$ at $650\ \text{nm}$. The index changes obtained with type I waveguides are volatile, since a thermal treatment at 150°C for 90 min erases the induced refractive index.¹²⁴

Type II: For $E \gg 0.5\ \mu\text{J}$, in x-cut samples, a strong material damage and a refractive index decrease in the laser focus

were shown. Additionally, light guiding regions appear at the sides of the damaged spot.¹²³ In contrast to the type I waveguides, these structures supported both polarizations but if the waveguide was inscribed in z-direction, no waveguide appears. Structures of type II are observed in x-cut substrates regardless of the direction of the inscribed line and the polarization direction of the writing laser, but cannot be formed in z-cut LiNbO_3 . These waveguides are stable upon annealing.

These experimental results evidenced that the mechanism responsible for the refractive index modification is not due to photorefractive effect and is different in type I and type II waveguides.¹²⁵ For the type I, the nonlinear absorption of the pulse strongly influences the spatial distribution of the energy delivered in the focus. When the laser intensity reaches the threshold for nonlinear absorption well before the focus position, the energy provided to the crystal in the focal point is reduced, resulting in only weak damage. This weak damage coincides with an increase in the extraordinary refractive index n_e , while no noticeable change of the ordinary refractive index n_o is observed in this case. These findings can be explained taking into account a partial amorphization mechanism, similar to the one reported in Section IV A, which lead to an increase in the extraordinary refractive index for moderate damaging.¹²⁵ In the case of type II waveguides instead, the high laser fluence produces a strong disorder in the crystal, resulting in an expansion of the crystal lattice. As a result, although n_o and n_e are decreased at the focal spot, the induced stress increases the refractive index in the surrounding material *via* elasto-optic effect. In this case, it is the compensation of the stress-induced densification in the focal area to be correlated with the refractive index modification,^{125,126} as observed also in other materials, e.g., Ti-doped sapphire.¹²⁷ Furthermore, type II structures were demonstrated to be thermally stable up to more than 300°C but, in general, the refractive index jump achieved was lower than that obtained by Ti-in diffusion or proton exchange, being estimated close to 8×10^{-4} .

C. Applications

Photorefractive (PR) laser-written waveguides were popular during the 90's, when a considerable effort was

dedicated to develop dynamically reconfigurable, 3D optical interconnections, which are of importance in all-optical signal processing and in more explorative applications, such as optical neural networks. The focus here was to have some connections dynamically reinforced or deleted according to some feedback action. The erasure of a waveguide written with the above-described method can be performed by sending a sufficiently intense signal beam (modification beam) through it.¹²⁸ However nowadays, it seems that the interest for this kind of applications has somehow faded out.

On the other hand, fs-laser written waveguides are presently receiving considerable interest thanks to the availability of fs laser systems and for the flexibility of this method, especially for what concerns type II waveguides. In this case, as the light is guided mainly in the undamaged region, the waveguides maintain the nonlinear optical properties of the material. Type-II writing can be further improved by using a double-line scheme, (see Figure 8): In this case, the optical confinement is observed in a high index region in between two lines with reduced refractive index.^{123,129} The lines can be inscribed consecutively or simultaneously.¹³⁰ Recently, three dimensional waveguide-waveguide interconnections,¹³¹ Y-splitters,¹²¹ and couplers¹³² were demonstrated, as well as 3D splitters and waveguide arrays of coupled waveguides^{124,133} were reported. Notably, the fs writing technique was exploited to get optical waveguides on periodically poled lithium niobate by the multiscan technique. Second harmonic generation experiments from a fundamental wavelength of 1567 nm were demonstrated with a conversion efficiency of $18\%W^{-1}$ (Refs. 134–136). SHG from 1064 nm radiation could also be obtained using birefringent phase matching, with a conversion efficiency that was reported as high as 49%.^{130,137} The fabrication of femto-second laser-induced, first-order waveguide Bragg gratings in lithium niobate in the low repetition rate regime was demonstrated as well.¹²⁹ In this case, the waveguides were structured periodically along the propagation direction to achieve narrowband reflections at wavelengths around 1550 nm. Additionally, electrodes are employed to allow for electro-optic tuning of the spectral response, demonstrating the wavelength control of the central reflection peak by applying

a static external electric field (maximum shift of the reflection peak of $\Delta\lambda = 625$ pm).

VI. SOLITON WAVEGUIDES

A. Self-induced optical waveguides

Under certain conditions, lithium niobate supports some phenomena which result in optical beams propagating without diffraction, with any need for a preexisting structure. It is therefore as if a waveguide was created in the material by the light beam itself. This kind of phenomena requires an optical nonlinearity, i.e., the refractive index has to be somehow locally modified by the propagating beam. Mathematically this can be described by a special type of nonlinear wave equation and the solutions corresponding to an optical beam propagating without diffraction are named “solitary waves” or “solitons.” In this context, we are dealing with spatial solitons, i.e., non-diffracting optical beams, in contrast to temporal solitons which are light pulses of finite temporal duration that propagate without dispersion.

Depending on the type of nonlinear optical phenomenon used to achieve the self-confinement, two important classes of optical solitons in LN were reported: (a) quadratic spatial solitons,¹³⁸ where the self-confinement is based on the power exchange between a fundamental wave and its second harmonic mediated by a quadratic nonlinearity; (b) photorefractive solitons¹³⁹ which rely on a combination of light induced charge transport and electro optic effect to produce the waveguide. In the following, we will discuss only the second type of solitons because they can be considered as an alternative way to produce a “real” waveguide, which remains written inside the sample and is capable of guiding a second optical signal. This approach presents several interesting advantages: It does not require processing the sample in any way, and is very easy and cost-effective compared to more traditional fabrication techniques. Moreover, waveguides can be realized in the bulk of a sample, opening the way for three-dimensional optical circuitry, and are self-aligned with the beam that created it which may be of interest for special applications. The disadvantages are that the obtained waveguides in general do not show a strong confinement.

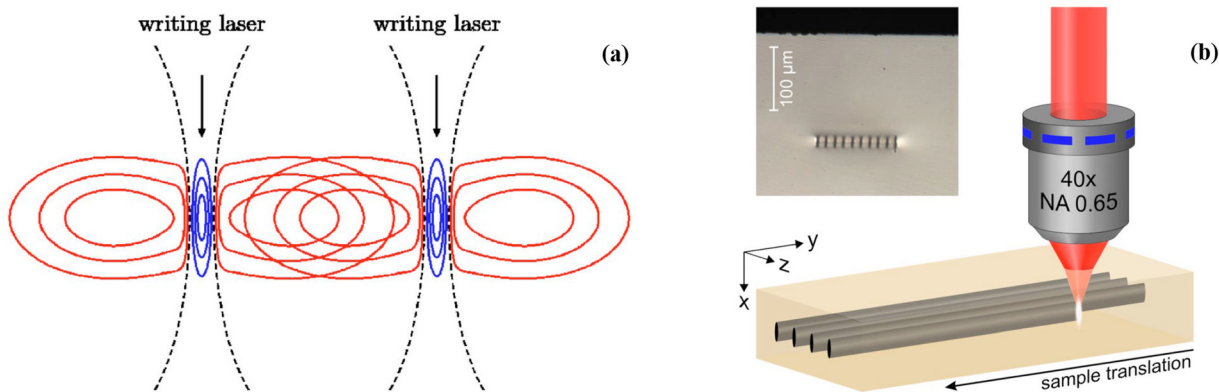


FIG. 8. Scheme of the type II waveguide fabrication technique: (a) Two lines were written so that a symmetric waveguide is formed in the centre (reproduced with permission from J. Burghoff *et al.*, Appl. Phys. Lett. **89**, 081108 (2006). Copyright 2006 American Institute of Physics. (b) Scheme of the apparatus and relative results (reproduced with permission from M. Heinrich *et al.*, Appl. Phys. Lett. **93**, 101111 (2008). Copyright 2008 American Institute of Physics.

Moreover, these optical structures can be erased by illuminating them with a strong light.

B. Overview of different photorefractive solitons in LN

In LN, solitons can be either “dark” or “bright.” In the first case, we are referring to a “lack” of light intensity in an otherwise uniform illumination pattern. Both types of solitons can be either one-dimensional (i.e., dark or bright light “stripes”) or two-dimensional (i.e., localized in both the transverse directions) and can be used to produce waveguides for a second beam. (1 + 1)D and (2 + 1)D dark solitons are readily obtainable simply by letting an appropriate optical beam to be transmitted through a properly oriented LN crystal.^{140,141} This is because LN shows a strong photovoltaic (photogalvanic) effect which, combined with the sign of the electro-optic tensor, naturally creates a self-defocusing of the optical beam, the necessary prerequisite for the formation of dark solitons. The waveguiding effect is obtained because in the outer bright regions, the refractive index is lowered by the defocusing nonlinearity, while in the dark central spot, it remains essentially unaffected, so that a guiding structure may result. Soliton waveguides formed using this approach are indicated as “photovoltaic” dark solitons. On the other hand, bright solitons requires a self-focusing, and this can be achieved only by some kind of external action, compensating the defocusing effect naturally present in LN. Generally, a bias electric field is applied in order to displace photoexcited charges in the correct direction to create a positive change of the refractive index. This second approach is named “photorefractive photovoltaic-screening soliton” because the above mentioned bias field results non-uniformly screened by photoconductive charges and contrasted by the photovoltaic field.¹⁴² The field can be applied along the crystal z axis (i) by making use of some external circuit;¹⁴³ (ii) by heating the crystal by some tens of degrees above room temperature, thus establishing a strong pyroelectric field in the bulk of the crystal;¹⁴⁴ (iii) by applying a uniform additional background illumination with an extraordinary polarization to the (ordinarily polarized) soliton beam¹⁴⁵ in an open-circuit configuration, so that a uniform electric field is created inside the crystal by the photovoltaic effect induced by the background light. These techniques allow obtaining qualitatively identical results, provided that the induced uniform field remains constant for the time required for soliton formation.

As the photorefractive soliton is a phenomenon triggered by light-induced charge transport phenomena inside the material, the defect structure of the sample plays a decisive role in the soliton formation process. Reduced samples are more photoconductive, and doping with photorefractive-enhancing ions like Fe or Cu may be used to increase the light induced refractive index changes. Finally, in literature, a distinction is generally made between quasi steady state and steady state solitons,¹³⁹ meaning in the first case that the soliton formation process is stable only for a limited time, so that the newly formed soliton beam experiences some instability which leads to filamentation and eventually to destruction of the waveguiding effect. In contrast, steady state

solitons are solutions which form over a typical time of few seconds depending on the beam intensity and remain stable for long times. To obtain this kind of effect, the dark conductivity of the sample must be artificially increased, e.g., by the additional use of some background illumination. From the fabrication point of view, it should be however mentioned that it is sufficient to stop the illumination pattern before the soliton begins to corrupt to have a usable waveguide, provided that the “signal” beam subsequently injected in the self-written waveguide has either a low power or a long wavelength, in order not to destroy the light-induced refractive index change.¹⁴⁶

C. Theoretical description and existence curve

The first theoretical descriptions of the propagation of an optical beam into a self-defocusing photorefractive crystal, with or without an applied field and eventually with additional illumination were provided in the 90's.^{118,142,147} For soliton formation, the (2 + 1)D problem can hardly be attacked by analytical methods, so that in general, theoretical treatments are limited to the simpler (1 + 1)D case. It can be shown, however, that the general behavior is qualitatively valid also in the higher dimensionality case.¹³⁹

The main idea is to set down a self-consistently coupled problem where the light intensity profile determines the changes in the refractive index which, in turn, modifies the light propagation in the medium. For LN, the photorefractive phenomenon is described in full generality by the Kukhtarev–Vinetskii equations.¹⁴⁸ These equations permit to link the illumination pattern $I(x)$ to the space charge field E_{PR} which sets up inside the material due to photoexcitation and migration of mobile charges. For typical (1 + 1)D experimental cases, these equations lead to a solution of the form¹³⁹

$$E \approx -(E_{PV} + E_0) \frac{I/I_{dark}}{1 + I/I_{dark}}, \quad (11)$$

where I_{dark} is the so-called *dark irradiance*, i.e., the light intensity that would produce a photoconductivity equal to the conductivity of the material in absence of illumination due to the thermal excitation of carriers. Because of electro-optic effect, a modulation of the refractive index is associated with the electric field E given by the following equation:

$$\Delta n = -\frac{1}{2} n_b^3 r E, \quad (12)$$

where n_b and r are, respectively, the material refractive index and electro-optic coefficient according to the chosen experimental conditions for the sample orientation, beam polarization, and so on.

Now, by substituting Eqs. (11) and (12) in the wave equation describing the propagation of an optical beam, and searching for a solution with a “solitonic” character (i.e., whose intensity is not dependent on z) of the form $A(x, z) = u(x) \exp(i\gamma z) I_{dark}^{1/2}$, two physically possible situations are obtained which are distinguished by the parameter $a = kn_b^2 r (E_{PV} + E_0)/2$, where $k = 2\pi/\lambda$:¹³⁹ (i) either $a < 0$ and $\gamma < 0$ (defocusing nonlinearity) or (ii) $a > 0$ and $\gamma > 0$

(focusing nonlinearity). The scale length of the problem is fixed by the real quantity $d = (\pm 2ka)^{-1/2}$ (where the “−” sign is chosen for $a < 0$ and vice versa), so that the transverse distance could be renormalized in an adimensional quantity: $\xi = x/d = x\sqrt{\pm 2ka}$. The following nonlinear differential equation is therefore obtained for the soliton amplitude $u(x)$:

$$\frac{\partial^2 u(\xi)}{\partial \xi^2} = \pm \left(\frac{\gamma}{a} - \frac{1}{1 + u(\xi)^2} \right) u(\xi), \quad (13)$$

where the “+” sign is for the $a > 0$ case (i.e., $\Delta n > 0$, focusing nonlinearity) and the “−” sign for $a < 0$ case (i.e., $\Delta n < 0$, defocusing nonlinearity). In the first case, the solutions are bright solitons, and dark solitons in latter. In particular for LN, in absence of external field, it is well known that a beam tends to defocus. It is instructive to consider typical values for this case:¹⁴⁷ In normally doped Fe:LN (Fe concentration lower than 0.1 mol%), the typical photovoltaic fields do not exceed 10^7 V/m and the maximum PR refractive index changes for extraordinarily polarized light range from 10^{-4} to 10^{-3} . For a wavelength $\lambda = 500$ nm and $n = 2.2$, this yields $d = 1/\sqrt{-2ka}$ comprised between 1.20 and $3.79 \mu\text{m}$. However, in accordance with Eqs. (11) and (12), by applying a field with a magnitude larger than E_{PV} and with opposed sign, a can be turned positive, allowing for bright solitons. In Ref. 143, the focusing is obtained using a field of the order of 10^6 V/m, which is consistent with the fact that the used sample was an undoped one.

In order to produce a soliton, one must launch a beam with a profile close enough to the solutions of Eq. (13). After a transient, the system will be attracted towards the stable state, i.e., the soliton. In the bright soliton case, these solutions have generally a bell-shaped profile and there exists a relationship between their FWHM and their peak amplitude which can be calculated using Eq. (13). This is a consequence of the saturable nature of the nonlinearity (Eq. (11)) which indicates that beams with different intensities give different soliton profiles. More precisely, I/I_{dark} plays a role in the phenomenon of self-trapping instead of the actual intensity profile $I(x)$. This gives another important tool to control the soliton formation process, because the dark irradiance can be artificially increased by putting some incoherent background illumination. From a practical point of view, this means that depending on the different experimental conditions (summarized by the parameter $a = kn_b^2 r(E_{PV} + E_0)/2$ and by the dark irradiance I_{dark}), for a given beam intensity, only beams with a specific full width at half maximum are likely to produce a soliton. Using Eq. (13), one can construct an “existence curve”¹³⁹ for bright or dark solitons which links together the experimental parameters of the input beam (normalized FWHM $\Delta\xi$ and normalized peak amplitude $u_0 = \sqrt{I_{\text{max}}/I_{\text{dark}}}$) for a given choice of the nonlinear parameter which depends only on material properties and on the applied field.

D. Experiments and characterization

In LN, the simplest soliton waveguide experiment concerns the formation of dark (1 + 1) D solitons which are

readily obtainable taking advantage of the natural photovoltaic defocusing of the material. Due to the anisotropic properties of the photovoltaic field, the refractive index gradient is preferentially aligned along the c axis. In Figure 9, shown is the spatial profile of a dark notch before and after the self-confinement, where the compensation of diffraction can be appreciated. The associated slab waveguide is shown in the lower part of the figure. Using specially produced beams, such as Laguerre–Gauss modes or other donut-like modes, (2 + 1) D dark solitons can be obtained (optical vortices).¹⁴⁹ In other experiments, by applying an electric field parallel to the c axis, it is possible to force the system to a self-focusing behavior (first row of Figure 10). The refractive index profile associated with a bright soliton waveguide was measured in Ref. 150 using an interferometric technique, with conditions close to those of Ref. 143. Those experimental results show that the phase map has a complex structure (Figure 11): The waveguide core is surrounded by two “lobes,” where the refractive index is lower than the bulk value. Interestingly, this situation is the exact opposite of what observed when a focused light beam is transmitted through LN without applied field (see Figure 7(a)). These situations are clearly beyond the simple model detailed in preceding paragraph (in particular, Eq. (11) does not hold), and this is due to the higher dimensionality of the problem which requires a more complicated analysis.¹¹⁸ Since the sample used in these experiments was an undoped one, the refractive index jump is quite modest, of the order of 10^{-5} . Larger refractive index contrasts can be achieved using appropriate doping with photorefractive enhancing ions, however in this case, the stronger confinement may de-stabilize the soliton, leading to bending and filamentation. Moreover, the doping generally increases the photovoltaic field of the

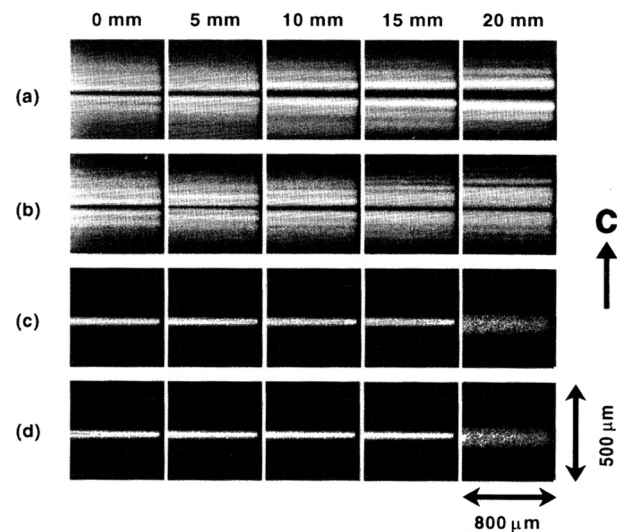


FIG. 9. Formation of a (1 + 1)D dark soliton and of the associated slab waveguide measured along the propagation distance (a) naturally diffracting dark notch. (b) Same after the formation of the dark soliton. (c) Co-propagating probe beam before the soliton formation. (d) Same after the formation of the soliton waveguide. The width of the input dark notch is about $15 \mu\text{m}$ and the writing beam has an average intensity of 10^5 W/m^2 from the 488 nm line of an Ar laser. The probe beam is the 514 nm line filtered from the same laser system. Reproduced with permission from M. Taya *et al.*, Phys. Rev. A **52**(4), 3095 (1995). Copyright 1995 The American Physical Society.

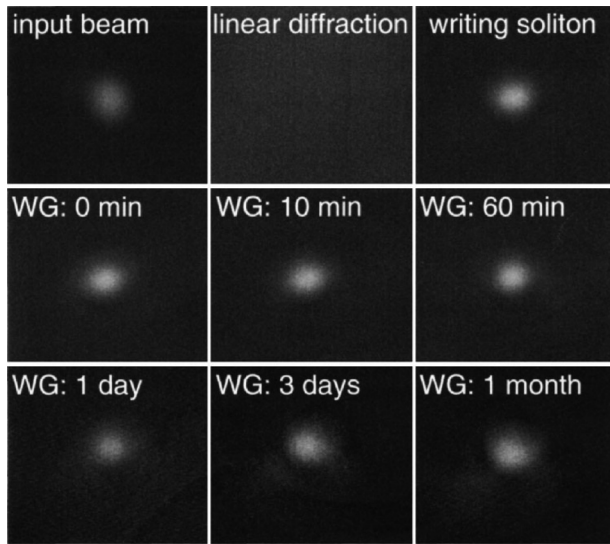


FIG. 10. Soliton formation process in an undoped congruent LN sample (first row) and resulting waveguide. $E_0 = 3.5 \times 10^6$ V/m, $\lambda = 514$ nm, $I_{\text{dark}} = 10$ W/m², $I_{\text{max}}/I_{\text{dark}} = 1500$. The obtained waveguide is stable up to one month (second and third row). Reproduced with permission from E. Fazio *et al.*, Appl. Phys. Lett. **85**(11), 2193 (2004). Copyright 2004 American Institute of Physics.

sample, so that in principle, much higher applied voltages are required to generate a bright soliton.

E. Applications

Up to our knowledge, a real working integrated optical device exploiting soliton technology in lithium niobate

seems not available yet, there exists a number of interesting proof-of concept applications which take advantage of the unique properties of self-written waveguides.

A non-exhaustive list is reported in Table III, where the selection is limited to experimental demonstrations. It should be mentioned that in other photorefractive materials, such as SBN or KNbO₃, other kind of applications have been demonstrated, and in principle, many of them could be reproduced also in LiNbO₃. For a list of these approaches, see, e.g., Ref. 139.

VII. WAVEGUIDES WITH PHOTOREFRACTIVE PROPERTIES

One of the interesting properties of LN in developing integrated optical circuits stems in photorefractivity as a way to provide new functionalities. Especially during the 90's, it was realized that this phenomenon, which was initially considered a limit in standard applications, could provide new opportunities to implement nonlinear interaction between two optical signals so that a considerable effort was devoted to exalt the photorefractive effects in a waveguide configuration.¹⁶⁰ The advantage of using a waveguide geometry compared to free-space photorefractive optics is that even modest light powers below the mW level can lead to high light intensities over long interaction distances. This results in fast photorefractive response time, on a timescale between the millisecond and the second. An alternative approach is to use the PR effect as a cheap and easy way to record photonic structures, such as, e.g., Bragg reflectors, along a conventionally fabricated waveguide. In the following, the main aspects

TABLE III. A list of experimentally proven applications of soliton waveguides in lithium niobate.

Type	Formation mechanism	External field conditions	Background illumination	Doping	Application	Notes	References
Dark (1 + 1)D	Photovoltaic	...	...	Undoped congruent	Slab waveguides	...	140
Dark (1 + 1)D	Photovoltaic	...	...	Fe-doped	Channel waveguides	Multiple exposures	151
Dark (2 + 1)D	Photovoltaic	...	...	Fe-doped	Channel waveguides	...	149
Dark (1 + 1)D	Photovoltaic	...	...	Undoped congruent	Slab Y-junction	...	152
Bright (2 + 1)D	Photovoltaic-screening	Electrical contacts	Incoherent light	Undoped congruent	Channel waveguides	Continuous and pulsed laser regimes	143 and 153
Bright (2 + 1)D	Photovoltaic-screening	Electrical contacts	Perpendicularly polarized laser beam	Undoped congruent	Sharp 90° bends by total internal reflection	Prism-like sample	154
Bright (2 + 1)D	Photovoltaic-screening	Electrical contacts	...	Er-doped	Luminescent waveguides	...	155
Bright (2 + 1)D	Photovoltaic-screening	Pyroelectric field	...	Undoped congruent	Integrated optofluidic refractive index sensor	Soliton beam propagating across a fluidic channel	156
Periodic	Photovoltaic	...	...	Undoped congruent	Parallel waveguide arrays	Simultaneous writing of a large number of waveguides	157
Dark (1 + 1)D	Photovoltaic	...	...	Fe-doped	Second harmonic generation	High intensity fs pulses confined by a dark soliton waveguide	158
(2 + 1)D	Photovoltaic-screening	Electrical contacts	...	Undoped congruent	Second harmonic generation	Bright soliton waveguide created by the converted signal, confining both fundamental and harmonic beams	159

of waveguide systems with enhanced photorefractive properties will be reviewed, for a more detailed treatment, the reader is referred to Ref. 160 (Table IV).

A. Fabrication

The standard Ti diffusion technology (Section II) is already sufficient to give measurable photorefractive sensitivity, especially at visible wavelengths. However, in order to obtain strong nonlinearities, the waveguides can be prepared on substrates bulk doped with ions enhancing the PR response, such as Fe or Cu. As an alternative, thermal co-diffusion processes can be used to dope the material and subsequently to fabricate the waveguide. As reported in Ref. 160 and references therein, the refractive index profile as a function of the depth for slab waveguides prepared in this way can be calculated as

$$\Delta n(z) = \Delta n_{\text{Ti}}(c_{\text{Ti}}(z)) + \Delta n_{\text{M}}(c_{\text{M}}(z)), \quad (14)$$

where $\Delta n_{\text{Ti}}(c_{\text{Ti}}(z))$ is the refractive index profile associated with the concentration distribution of Ti atoms, and $\Delta n_{\text{M}}(c_{\text{M}}(z))$ is the same for the concentration profile of PR enhancing dopant (M = Fe, Cu, in our case). Proton exchange waveguides and annealed-proton exchange waveguides (Section III) have generally a small PR sensitivity, due to a large dark conductivity and smaller electro optic coefficients compared to Ti in-diffused waveguides. However, by using a second ion-exchange treatment,¹⁶⁰ the crystal surfaces can be doped with photorefractive metallic ions, such as Cu. By perfecting the technique with suitable annealing treatments, it is possible to obtain waveguides with low losses, high optical confinement, and good PR sensitivities. Waveguides prepared by ion implantation over a LN substrate doped with PR enhancing ions are another possibility. Implantation of light ions, such as H or He, generally shows a degradation of PR properties compared to bulk material, due to the reduction of the electro-optic coefficients as a consequence of the high fluences required. The situation is more favorable by using heavier ions like oxygen: In this case, the photorefractive sensitivity is practically equal to that of the substrate.¹⁶¹ Also in this case, doping with Fe or Cu can be used to obtain strong photorefractive sensitivities. The PR characteristics of the produced waveguides depend on the local concentration of dopants and on their chemical reduction degree, which is defined in a local fashion as the concentration of donors over that of acceptors.¹⁶² These parameters are determined by (i) the doping parameters; (ii) the

thermal treatment; and (iii) the substrate type.^{163,164} It should be pointed out that these characteristics may change along the waveguide.¹⁶² These facts may explain a certain scattering in published data.

B. Characterization of the PR properties

The photorefractive properties of a waveguide can be evidenced quite straightforwardly in a slab configuration. In this case, three classes of methods can be used (see Figure (12)): (i) by studying the fanning out of a single extraordinarily polarized signal beam which occurs if the optical axis c lies in the slab plane; (ii) by performing a photorefractive wave mixing between two signal beams which crosses inside the slab; (iii) using some external optical illumination (e.g., a sinusoidal light pattern) to alter the propagation of a guided signal beam. The maximum sensitivity is obtained for extraordinarily polarized beams and if the scattering wavevector $\mathbf{K} = \mathbf{k}_2 - \mathbf{k}_1$ is parallel to the crystal c axis, where $\mathbf{k}_{1,2}$ are the wavevectors of the two writing beams, here approximated as two plane waves. However, different combinations of propagation directions, polarizations, and mode numbers can be used.¹⁶⁷

Concerning the characterization of channel waveguides, some older literature considers also the possibility to measure the PR properties by placing the device in one of the arms of an interferometric setup, which may be in free space or even integrated directly in the substrate.¹⁶⁸ It is also possible to directly detect light induced voltage differences and photocurrents depositing a couple of electrodes at the borders of the waveguide.¹⁶⁹ Some other methods^{170,171} were developed to analyze the PR properties of a waveguide with spatial resolution using micro-Raman spectrometry, while in Ref. 159, a series of micron-sized hologram is written using tightly focused laser beams on different points of the sample. By recording for each of them, the writing curve, the photorefractive parameters can be measured as a function of the position.

It is important to note that characterization methods based on diffraction from light-induced gratings can be significantly inaccurate if proper care is not devoted to data interpretation. Generally, in free space photorefractive optics, the amplitude of the refractive index modulation Δn is calculated from the intensities of the diffracted I_S and transmitted I_R beams measured by two photodiodes as depicted in Figure 12 using the Kogelnik's formula

$$\eta = \frac{I_S}{I_S + I_R} = \sin^2\left(\frac{\pi \Delta n d}{\lambda \cos \psi}\right), \quad (15)$$

TABLE IV. Maximum refractive index change Δn_{max} and photorefractive sensitivity $S = \Delta n/E$, with Δn is the refractive index change obtained after the absorption of E optical energy, for Ti in-diffused (TI), proton exchanged (PE), and Annealed Proton Exchanged (APE) waveguides.

Substrate	Waveguide type	Polarization	Δn_{max}	S (cm ² /J)	I (W/cm ²)	References
LN	TI	TE	2×10^{-4}	4×10^{-7}	10^3	165
	TI	TM	6×10^{-4}	9×10^{-6}	10^3	...
	PE	TM	4.2×10^{-6}	3.8×10^{-12}	1.3×10^4	166
	APE	TM	5.1×10^{-6}	5×10^{-9}	2.2×10^2	...
Mg:LN 7%	PE	TM	4.9×10^{-6}	1.8×10^{-11}	1.8×10^4	...
	APE	TM	5×10^{-6}	1.4×10^{-8}	10^2	...

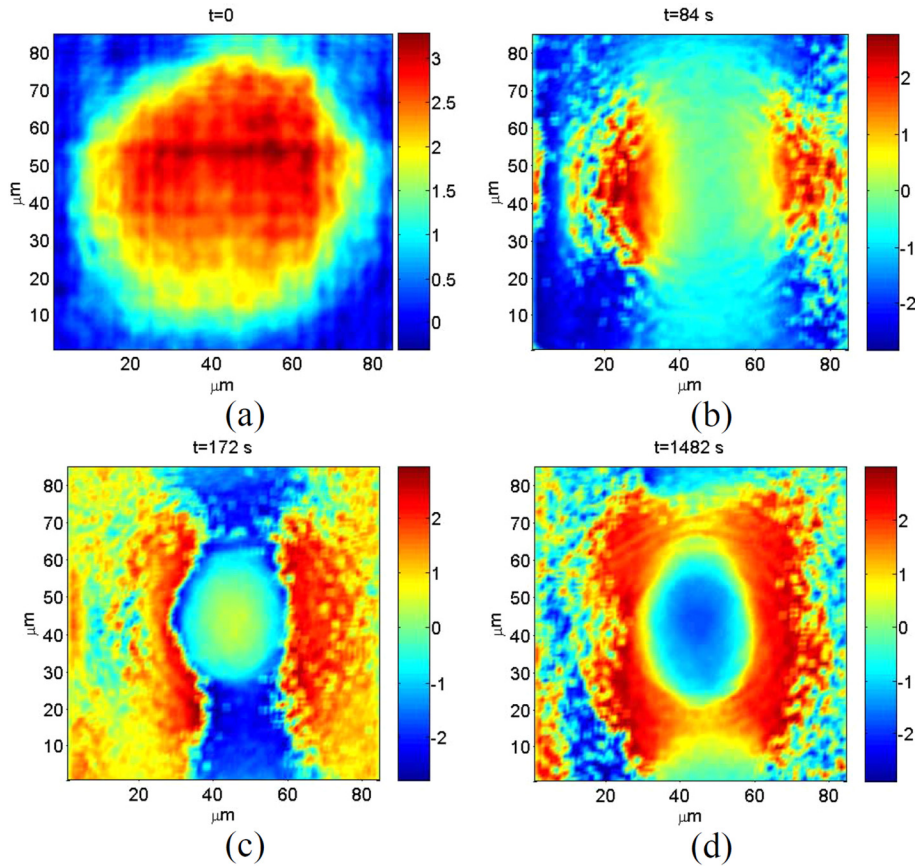


FIG. 11. Phase map of a bright soliton formation process after 10 mm propagation in an undoped congruent LN crystal. $E_0 = 3.5 \times 10^6$ V/m, $\lambda = 514$ nm, $I_{max} = 2.13 \times 10^4$ W/m², $I_{dark} = 12$ W/m². The c axis is parallel to the horizontal axis of the figures. Reproduced with permission from M. Paturzo *et al.*, Opt. Express **15**, 8243 (2007). Copyright 2007 Optical Society of America Publishing.

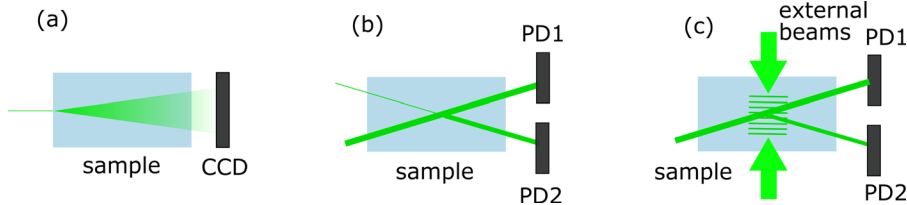


FIG. 12. Different configurations for testing the photorefractive sensitivity of a slab waveguide. (a) Beam fanning method; (b) wave mixing method; and (c) externally written grating method.

where d is the thickness of the grating, λ is the signal wavelength, and ψ is the angle between the propagation direction of the transmitted beam and the normal to the sample surface. However, as pointed out in Ref. 172, to get reliable results, it is important to take into account of the fact that in a waveguide, the beam depends also on the transverse coordinates because of the waveguide confinement. Moreover, as the power density in waveguides is generally larger than in free space, it is necessary to take into account also two-step phenomena, which are due to the interplay between dopant impurities (Fe or Cu) and intrinsic PR centers. In this situation, the photorefractive quantities depend on the intensity, and therefore it is necessary to test different waveguides using approximately the same beam intensity to make a reliable comparison. An updated extension of the Kukhtarev–Vinetskii model for photorefractivity to this situation can be found in Ref. 173.

C. Applications

By combining photorefractive grating writing with thermal fixing recipes, it is possible to get permanent optical

structures. In Ref. 160, this approach was used to produce interference filters with a stop band of about 0.1 nm at telecom wavelengths and with a reflectivity of 60%. These structures can be used also as wavelength-selective mirrors or distributed-Bragg reflectors useful, for instance, in the fabrication of integrated laser sources in LN (Refs. 174 and 175) (see Section II C).

Because of the negative photorefractive nonlinearity of LN, by exploiting a suitable illumination system, it is possible to create optical barriers that, if properly arranged, resulted in a waveguide. The illumination generally is shone on the top of the sample, so that only the lateral confinement could be achieved. Therefore, this approach is generally applied to a preexisting slab waveguide structure, which provides the vertical confinement (see Figure 13).

The slab waveguide was produced by ion implantation of different species^{176,177} or by Ti in-diffusion.¹⁷⁸ The illumination can be obtained either by depositing a photolithographic mask as in Figure 13 or applying a uniform incoherent illumination from the top of the sample main surface. Alternatively, some type of structured illumination can be produced, like in Ref. 178, where a dark notch was used,

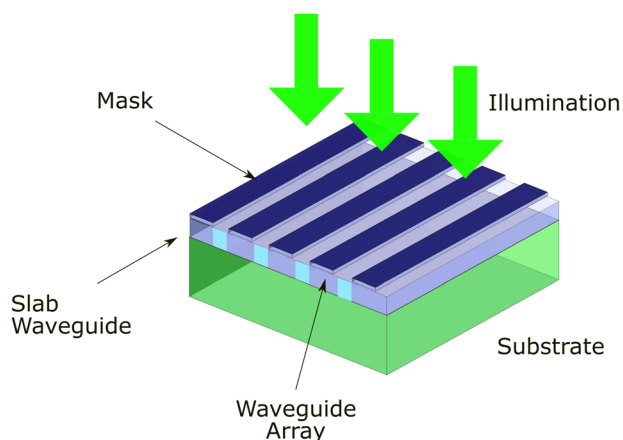


FIG. 13. Exemplary scheme for the fabrication of static waveguide arrays by structured illumination.

as in 1-D soliton experiments, Section VID. Some authors reported also the fabrication of 3D light induced waveguide arrays by sequential exposure to the interference patterns of two plane wave-like beams¹⁷⁹ crossing in the volume of the sample¹⁸⁰ or other type of structured illumination. In this case, no pre-existing waveguide is needed. The samples are generally doped with Fe, in order to enhance the photorefractive sensitivity of the material, with the drawback of an increased absorption at visible wavelengths. The refractive index contrast that can be expected with these approaches is similar to the one obtained in soliton waveguides, i.e., not much larger than 10^{-3} even for strong Fe doping (0.1 mol. %). Even if some basic functionality was demonstrated with this system, namely, Y-branches¹⁷⁷ and waveguide coupling, to our knowledge, no technological applications beside academic research have been reported with this fabrication technique. On the other hand, the unique properties of reconfigurable waveguides offer a concrete advantage whenever rapid prototyping or fabrication of large waveguide arrays are required and can be used for the demonstration of interesting new optical phenomena.^{181,182}

Photorefractive waveguide allows a unique way to perform all-optical beam interactions. Compared to bulk crystals, the quantization of waveguide modes and the inhomogeneity of the light field due to internal scattering may lead to quite peculiar processes, such as self-scattering (see Figure 12(a)), intermode coupling, and coupling between orthogonally polarized modes.¹⁶⁰ All these phenomena are the basis for the realization of integrated optical devices in LN based on the concept of photorefractive wave mixing. Using PR beam coupling, large amplification of weak optical signals is possible even at moderate powers¹⁸³ and was recently used to demonstrate some exotic phenomena, such as the violation of the time-parity symmetry¹⁸⁴ in an optical system. Self-aligning grating couplers were fabricated in Ti diffused waveguides prepared on Fe:LN substrates by letting to interfere a guided beam with a free propagating one.¹⁸⁵ Phase conjugation, i.e., the production of a time-inverted replica of an optical field can be implemented in a waveguide configuration using different geometries and can be applied to realize self-aligning integrated

optical cavities and auto-adapting beam couplers. In this case, it is useful to employ anisotropic wave mixing because in this way, the nonlinear interaction affects only the signal beam without perturbing the two pump beams, as demonstrated in Ref. 186. In this work, the authors showed a PR gain higher than one for the phase-conjugate wave in a planar waveguide fabricated by Ti and Fe co-diffusion.

One of the main interests in PR waveguides stems in the possibility for optical reconfiguration in order to achieve dynamic all-optical routing. This approach was proposed as the basis for implementation of complex functionalities, such as parallel vector-matrix multiplication.¹⁸⁷ The input and output vectors correspond to an array of waveguides, while the matrix operation is performed by a holographic grating, coupling the light pattern from the input array to the output one. The grating can be written using an external light pattern, such as, e.g., the one produced by a spatial light modulator.

In recent times, a growing interest in the scientific community was demonstrated for the use of integrated optics to realize optical analogues of quantum phenomena. This may serve both to prove experimentally some phenomena which are difficult to be observed in quantum mechanics but also to develop new optical system. Here, the analogy derives from the observation that the time-dependent Schrodinger equation has the same structure of a space-dependent Helmholtz equation, where the time coordinate is replaced by spatial propagation, time frequencies by spatial frequencies, and so on.¹⁸⁸ In this context, a waveguide corresponds to a potential well and the light field to a trapped “particle.” Optical coupling between waveguides describes the transition probability of the optical “particle” to other states. In this context, photorefractive waveguide, generally produced by ion implantation or Ti-diffusion in Fe:LN or Cu:LN substrates, has emerged as an excellent playground because they allow mimicking a nonlinear interaction between the different states. In fact, in coupled PR waveguide arrays, the coupling coefficient depends on the light intensity.

This was used to study the nonlinear behavior of optical waves distributed among different waveguides which shows some peculiar change in the propagation dynamics depending on the mutual phase and the intensity of the input beams.^{189–191} For example, in Figure 14, it is shown how two in-phase beams, which for low intensity become distributed among different waveguides along the propagation distance (discrete diffraction), can remain confined on their input channel or even oscillate between a restricted number of channels (gap solitons) according to the input intensity. These findings can be useful to develop new ways to perform optical routing where the output of a waveguide array can be changed only by adjusting power, but also to investigate diffusion of quantum particles along particular types of lattice. For example, in Ref. 192, an array of coupled waveguide in Fe:LN is used to simulate the equivalent of phonon-assisted Rabi Oscillation, a phenomenon which was theoretically previewed in other branches of physics but not yet observed. In this case, the photorefractive properties of the substrate were employed to record an additional refractive index modulation superimposed to that of the Ti-indiffused waveguide

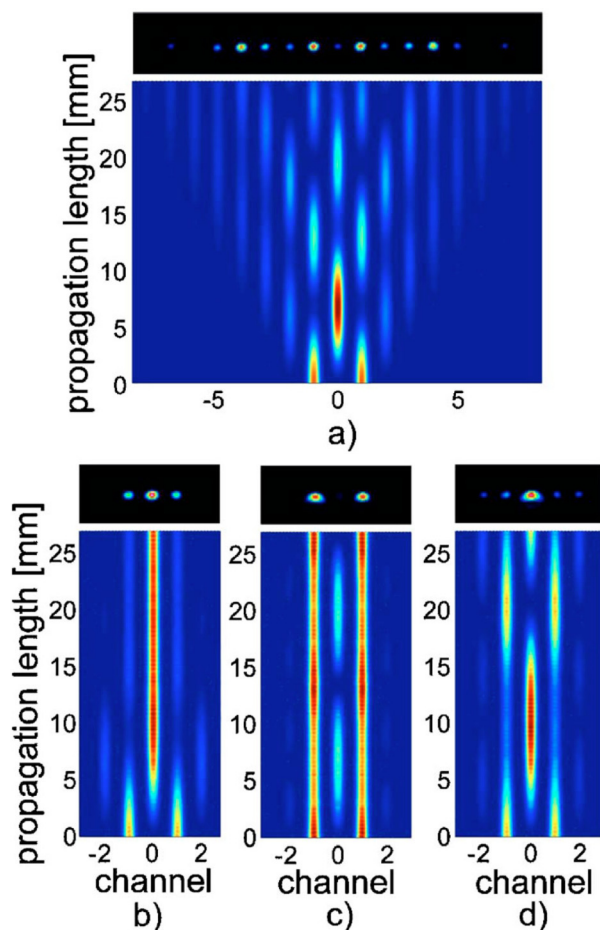


FIG. 14. Different coupling and propagation regimes between an array of parallel photorefractive waveguides attained for different input powers, at the output face (experimental), and along the propagation distance (theoretical). The two input beams are in-phase with each other. (a) Low power, discrete diffraction; (b) $I = 0.5 \mu\text{W}$, fusion of the two input beams; (c) $I = 7 \mu\text{W}$, beam separation; (d) $I = 30 \mu\text{W}$, coupled oscillations without discrete diffraction. Reproduced with permission from M. Stepić *et al.*, Phys. Rev. E **74**, 046614 (2006). Copyright 2006 The American Physical Society.

arrays, in order to simulate a phonon-assisted coupling between the different states of the quantum system. Other exotic phenomena are under study, such as the transition from spatial frequency combs to full delocalization of the optical wave along all the waveguide array which can be obtained when two optical Floquet-Bloch waves are made to interact via the photorefractive two-beam coupling and “special” transverse scattering vectors are selected by changing the propagation direction of the two input modes.¹⁹³ Interestingly, it has been shown that under the high intensity obtainable with fs laser pulses, the photorefractive properties of a Fe:LN waveguide array can cooperate with other nonlinearities and influence also spectral characteristics, such as dispersion and pulse duration.¹⁹⁴

VIII. SMART-CUT AND SUSPENDED WAVEGUIDES

These fabrication techniques, a by-product of the ion implantation technology, were derived by the semiconductor industry. They consist in detaching a high quality single crystalline film from a sacrificial wafer and use it as high-

index contrast waveguide, either standing free or bonded on the top of another substrate. In the latter case, a thin layer of material with a lower index is sandwiched between the LN film and the substrate, in order to create the needed optical barrier. Using this approach, it is possible to realize slab waveguides with superior optical and structural properties with respect to other bottom-up approaches, such as epitaxial growth, chemical vapor deposition, pulsed laser deposition, sol-gel, and so on.

This technology was pioneered by the Osgood’s group at the end of the 90’s¹⁹⁵ with the name of Crystal Ion Slicing (CIS). A single crystal LN wafer oriented along the *c* axis was implanted with He ions with a fluence of 5×10^{16} at/cm² and an energy of 3.8 MeV. This created a buried amorphous layer at a depth of about 10 μm below the surface, which could be dissolved by diluted hydrofluoric acid at a rate significantly larger than that of the undamaged material. After several hours of etching, a free-standing LN single crystal membrane with a thickness equal to the implantation depth was obtained. The residual damage provoked by the ion implantation process could be removed by a rapid thermal annealing process, so that at the end, the produced slab had linear and nonlinear optical properties close to the one of a bulk substrate and remained single-domain with the same orientation of the starting wafer.^{195,196}

This approach could be straightforwardly adapted for the production of Lithium Niobate On Insulator (LNOI) waveguides. The necessary steps are summarized in Figure 15.

Using this approach, LN smart cut waveguides with thicknesses below 1 μm could be obtained by bonding the LN film over suitably chosen substrates.¹⁹⁷ In this case, the implantation energy has to be reduced to few hundreds of keV. The bonding process is performed by simply pressing together the two wafers after careful polishing and cleaning (Figure 15(b)) and heating at 220 °C for several hours. The problem of the thermal expansion matching between the LN thin film and the underlying structure can be easily solved by using a LN substrate as supporting medium and keeping the thickness of the sandwiched insulating layer (generally SiO₂) below a certain critical value (about 1 μm). However, recently, also the bonding of LN to Si wafers has been demonstrated.¹⁹⁸ Interestingly, with such an approach, a metal film can be included in the structure below the SiO₂ insulating layer: This offers a very convenient way to pilot the refractive index of the smart-cut waveguide by depositing a second electrode on the top of the LN film and applying an electric field between the two. In this way, it was verified the unprecedented result that the electro-optic properties of the thin LN waveguide are practically equal to those of the starting wafer.¹⁹⁷

By combining the implantation-induced amorphization and etching with other microfabrication techniques, such as FIB milling,¹⁹⁹ it is possible to realize also LN – On-Nothing structures, i.e., thin LN slabs with a cavity below (Figure 16).

This gives a very large refractive index contrast also in the vertical direction, enabling the fabrication of waveguides with strong confinement. Obviously, the same fabrication techniques allow also the lateral structuration of the smart cut and suspended structures, so that a number of different

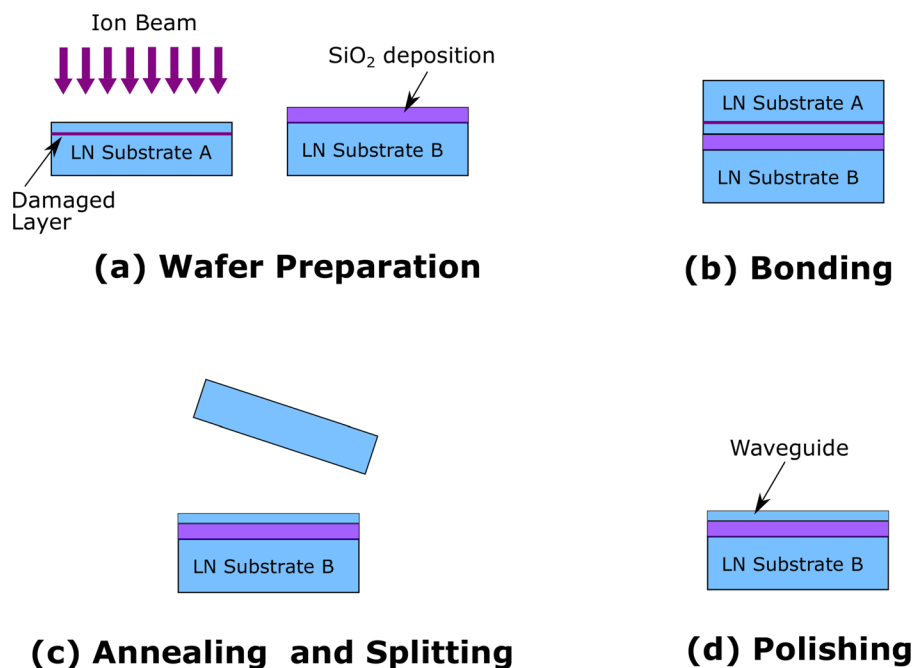


FIG. 15. Smart-cut technology.

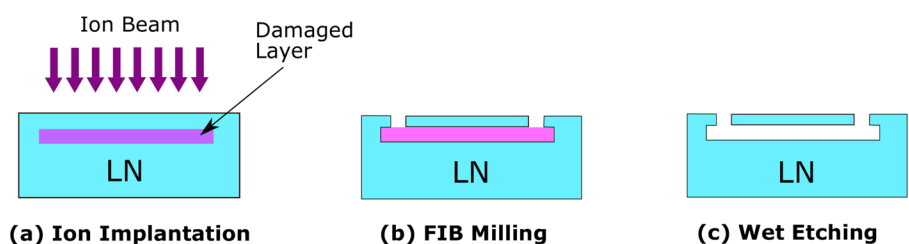


FIG. 16. Fabrication of a suspended waveguide.

photonic devices were demonstrated (see below). The advantages of this approach are the high refractive index contrast obtainable and the high quality of the waveguides, which present very low losses and almost the same linear and non-linear properties of bulk LN. Moreover, this technology is compatible with periodic poling and doping and facilitates the design of hybrid semiconductor-LN devices, a key point in the development of the next-generation optical processors. As a consequence, it is likely to play a major role in the future development of LN-based photonics. Indeed, in spite

of the fact that this approach is relatively recent, 3-in. wafers of LNOI are already commercially available.²⁰⁰

A. Fabrication examples and typical performances

High confinement ridge waveguides were fabricated using smart-cut slab waveguides and lateral ion milling techniques with a lithographically defined mask (Figure 17).²⁰¹ The thickness of the smart-cut waveguide was 760 nm and the lateral width was reduced up to 1 μm . This allowed for a

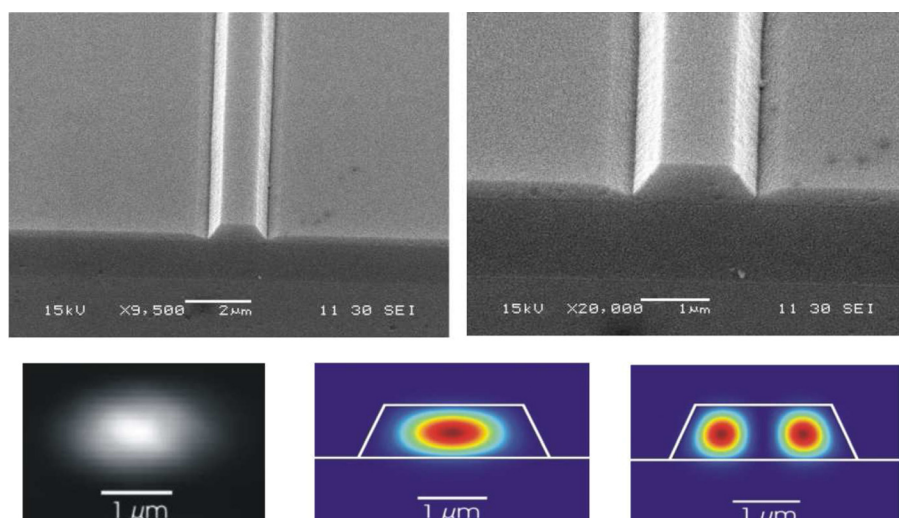


FIG. 17. (Top) SEM image of a 1 μm -wide ridge waveguide prepared on a smart-cut LNOI substrate. (Bottom) Measured near-field intensity distribution compared with calculated TM_{00} and TM_{01} waveguide modes. Reproduced with permission from H. Hu *et al.*, Opt. Express **17**, 24261 (2009). Copyright 2009 Optical Society of America Publishing.

mode field area as small as $0.4 \mu\text{m}^2$ at $1.5 \mu\text{m}$ wavelength, i.e., about one order of magnitude smaller than conventional Ti-diffused waveguides. This is of great interest for nonlinear optical application, as it allows obtaining high power densities even at low input powers. The optical losses were of the order of 10 dB/cm for both polarizations: a very good value when the extreme confinement of the field is taken into account. Using the same technology, photonic structures allowing for the realization of “slow light” schemes can be implemented.²⁰² Here, the idea is to artificially increase the effective index of the medium by exploiting an engineered photonic band structure; in this way, the light can “spend more time” inside the modulator arm so that the quarter-wave voltage is lowered of many times. The photonic band structure consists of an hexagonal arrays of air holes drilled on the smart cut guide by Ar milling, with a waveguiding structure of missing holes. The same concept is applied to a suspended LN waveguide,¹⁹⁹ where the higher vertical confinement allowed for an even better control of the dispersion properties of the structure (Figure 18).

Another intriguing possibility enabled by the smart-cut technology is the realization of periodically poled lithium niobate (PPLN) waveguides. First approaches in this respect explored the possibility of performing a poling process directly on a c-oriented smart cut waveguide with a metal electrode embedded either below the insulating SiO_2 bottom layer or between the insulating layer and the LN smart cut film.²⁰³ To this aim, a second electrode is patterned on the top of the waveguide by standard optical lithography and a voltage is applied between the two electrodes in order to induce a spatially patterned domain nucleation. Another possibility is to use a PPLN substrate (nowadays readily available) as the starting wafer from which to extract the PPLN

waveguide using the smart-cut process.²⁰⁴ Although the obtained results up to now are still not satisfactory (the resulting domain structures have not the required perfection so that the conversion efficiency is low), it seems that the full maturation of this process is still to be obtained. In spite of this, better results appear realistically at reach.

Alternatively, instead of using engineered domain structures, it is possible to exploit the unique possibilities offered by these advanced microstructuration techniques to build resonant photonic cavities, amplifying the nonlinear interaction between the fundamental and the second harmonic fields. The conversion efficiency can be therefore increased in order to compensate for the non-ideal phase matching. This approach was recently demonstrated in Ref. 205: A vertical coupling cavity was engineered on a suspended membrane of LN with a carefully designed photonic structure made of air holes. The excellent refractive index contrast allows for a high Q factor of about 500 at telecom wavelengths so that measurable second harmonic generation was observed even with a continuous pump beam of intensity as low as some μW . These performances are competitive with results obtained in III-V semiconductor nonlinear photonic structures, but with the advantage of a much larger transparency window offered by LN.

IX. PERIODICALLY POLED LN WAVEGUIDES

One of the most useful properties of lithium niobate is its large optical nonlinearity, which combined with the opportunity for domain engineering (see the review by Shur in this same series), allows for the fabrication of nonlinear optical components based on the Quasi Phase Matching scheme for efficient wavelength conversion. The waveguide geometry is particularly suited to this goal, because it allows for long interaction distances and high intensity levels even with moderate powers. The theoretical treatment of PPLN waveguides as well as their fabrication technology was developed between the 80's and the 90's. The main problem to solve was the interaction between the domain structures and the main waveguide fabrication processes used at those times (Ti-in diffusion and proton exchange), especially when the material was doped with photorefractive enhancing ions (such as Mg) to prevent the photorefractive effect.

Here, we limit to observe that the research was indeed successful and PPLN waveguides can now be fabricated by Ti in-diffusion, proton exchange, ion implantation, and more recently with fs-laser written and smart-cut waveguides, as reported in Sections IIC, IIIB, IVB, VC, and VIIIA. Some commercial devices based on PPLN waveguides are now becoming available,²⁰⁶ witnessing the fact that this technology is fully mature. The interested reader is referred to the works by the Fejer's group at the Stanford University²⁰⁷ for more details. Here, we will focus on most recent applications enabled by this kind of systems.

A. Applications

The classical application of PPLN waveguides is the realization of efficient sources in spectral ranges difficult to reach by direct laser emission. The production of visible

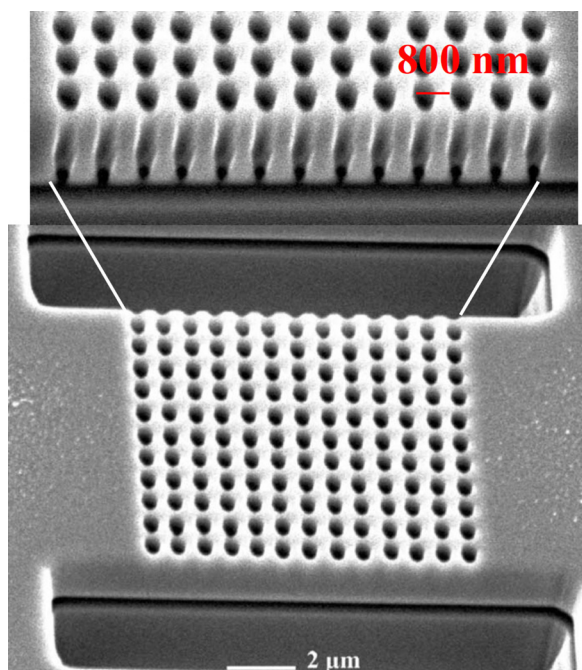


FIG. 18. Suspended LN slab waveguide with a Focused Ion Beam (FIB)-milled photonic structure. Reproduced with permission from G. Si *et al.*, J. Vac. Sci. Technol., B 28(2), 316 (2010). Copyright 2010 American Vacuum Society, AIP.

light using these systems seems today to be quite abandoned, owing to the competition of new semiconductor-based lasers and of other nonlinear optical materials. Instead, PPLN waveguides are well suited for the generation of mid-infrared radiation for free-space optical telecommunication²⁰⁸ and gas absorption spectroscopy.²⁰⁹ In both these papers, the authors employ Ti-diffused ridge PPLN waveguides to generate radiation in the 3.4–3.8 μm range by difference frequency generation because, contrary to proton exchanged waveguides, Ti:PPLN waveguides have nearly no absorption around 2700 nm wavelength due to the excitation of OH-vibrations. Efficiencies of more than 10%/W, resulting in several mW of converted radiation, were demonstrated. The same device can be used in the reverse way to convert long-wavelength light to a spectral range, where highly sensitive semiconductor detectors may operate, allowing for efficient detection of mid-IR light.²¹⁰ Thanks to the advances in LN fabrication technology, quite sophisticated development of this concept is now feasible, such as Integrated optical parametric oscillators (IOPOs) with built-in (intracavity) components like electro- and acousto-optical tunable filters, phase and amplitude modulators, Bragg reflectors as wavelength selective mirrors, wavelength-dependent couplers to get separate cavities for signal and idler, (tunable) pump lasers in the same structure, etc.²¹¹

Another intriguing application is the use of PPLN waveguides as logical elements in the realization of all-optical signal processing systems.²¹² In this case, the waveguide is fed with two optical beams, a signal and a pump, with different wavelengths. Exploiting frequency mixing, it is possible to generate a third signal with a new wavelength only if both the carrier beam and the pump are on. Optical switching in a PPLN waveguide has several attractive properties, including (1) quantum-limited spontaneous emission noise; (2) no added chirp; (3) wide operational bandwidth (>70 nm); (4) ultrahigh-speed operation (>1 THz); (5) similar up- and down-conversion efficiencies; and (6) multichannel operation with minimal crosstalk. In this case, proton exchanged waveguides can be exploited because of their ease of fabrication. From a system designer's perspective, a single-pump PPLN waveguide can therefore operate as an optically controlled multi-pole single-throw switch. This means that there can be several inputs, which are individually transferred to their corresponding outputs, when the optical control signal (the PPLN waveguide's pump) turns on. Using this element, it is possible, for example, to implement all-optical logical operations, such as AND and XOR (Figure 19) together with a number of other different functionalities for ultrafast data processing.²¹²

A further useful property of PPLN waveguides is their phase preserving nature; the output signal retains the phase properties of the input signal while introducing negligible optical SNR (OSNR) degradation. This allows SHG and wavelength conversion that can be controlled at bit-time scales without destruction of phase information. One way to exploit this aspect is for the compensation of signal degradation due to chromatic dispersion in long optical fiber links and/or spectral crosstalks.²¹² In fact, it is possible to build a phase conjugator system based on a nonlinear process taking

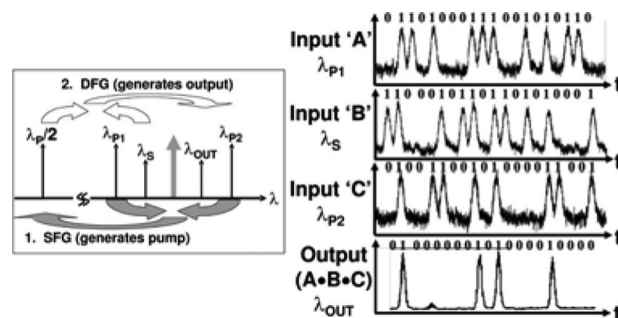


FIG. 19. Operation of a three-input AND gate. SFG between two of the inputs generates a product at the PPLN pump wavelength, which then mixes with the third input via Difference Frequency Generation (DFG), leading to a wavelength-converted output that represents the logic AND of all three inputs. Reproduced with permission from C. Langrock *et al.*, J. Lightwave Technol. 24(7), 2579 (2006). Copyright 2006 Institute of Electrical and Electronic Engineers.

place in the PPLN waveguide, so that if an input spectrum made of different wavelengths is injected in the waveguide together with a pump signal, the output spectrum is inverted with respect to the pump wavelength by difference frequency generation.²¹³ Interestingly, the inverted spectrum is also phase conjugated to the input one, so that if the PPLN phase conjugator is placed midway in the optical link, the optical signal will be restored after travelling through the second half of the optical fiber. Other applications of this kind regard encoding and processing of phase signals, for example, for new protocols of secure communications.²¹⁴

Finally, the rapidly increasing field of quantum information has found PPLN waveguides an essential element for the development of integrated demonstrators working at the single photon level.²¹⁵ They can be used not only to produce heralded single photon sources by production of entangled pairs *via* parametric down conversion^{215,216} but also to pilot the produced qubits exploiting the other functional properties of LN, such as electro-optic modulation.³⁷ The possibility opened up by waveguide technology in LN also comprehend direct implementation of quantum tests, among all the possibility to use couplers and Mach-Zehnder interferometers to design on-chip single photon interference experiments.²¹⁷

X. CONCLUSIONS

Lithium niobate is a material with incredible versatility. It is true that none of its characteristics shows record values, but the fact that all of them coexist in the same material is of paramount practical importance for the fabrication of multifunctional integrated devices. Photonics technology in LN has been developing since 50 years without stopping, producing a great variety of different results and applications, sometimes even un-expected. It looks now that the research is entering into a new phase, i.e., the transition from the realization of single building blocks to the implementation of complex systems into the same chip connected by waveguides with tailored properties that can even have interesting applications in other fields, such as opto-microfluidics and optical sensing. This process resembles the advent of integrated microcircuitry in silicon about 70 years ago which led to the tremendous development of the microelectronics and

the information technology. Probably, we are at the beginning of a new revolution, driven by today's challenges, such as ultra-fast data transmission and processing, secure communication, and all-optical computing and it is likely that LN photonic technology will occupy a key role in this field.

ACKNOWLEDGMENTS

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